

# Some “bad” visualization examples

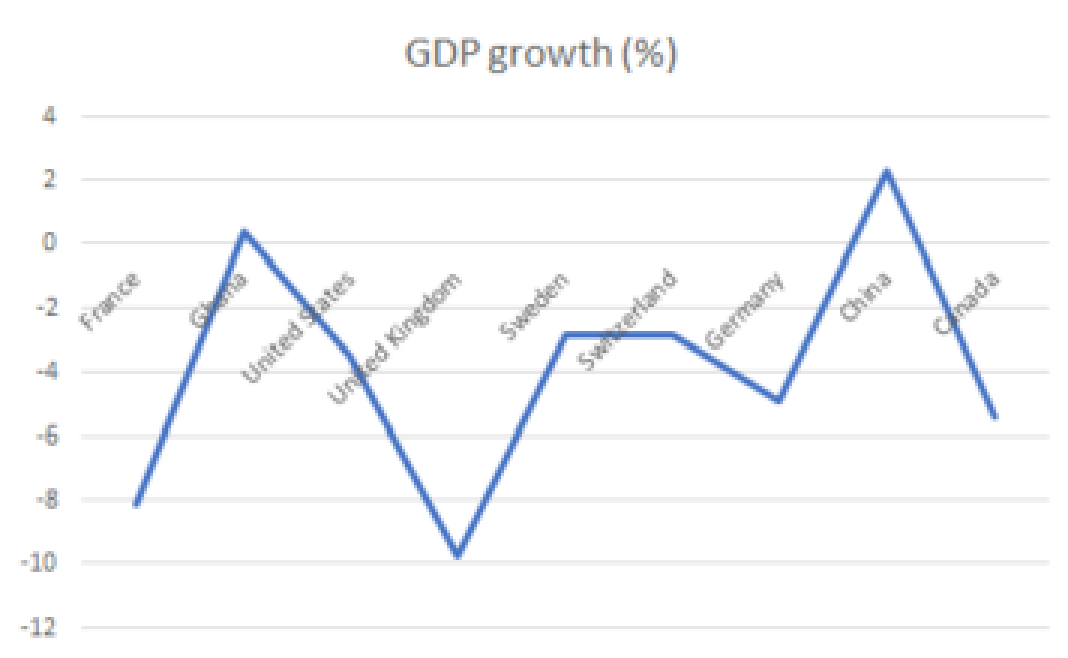
CT II – Visualização de Dados

Joaquim Madeira

# The Basics

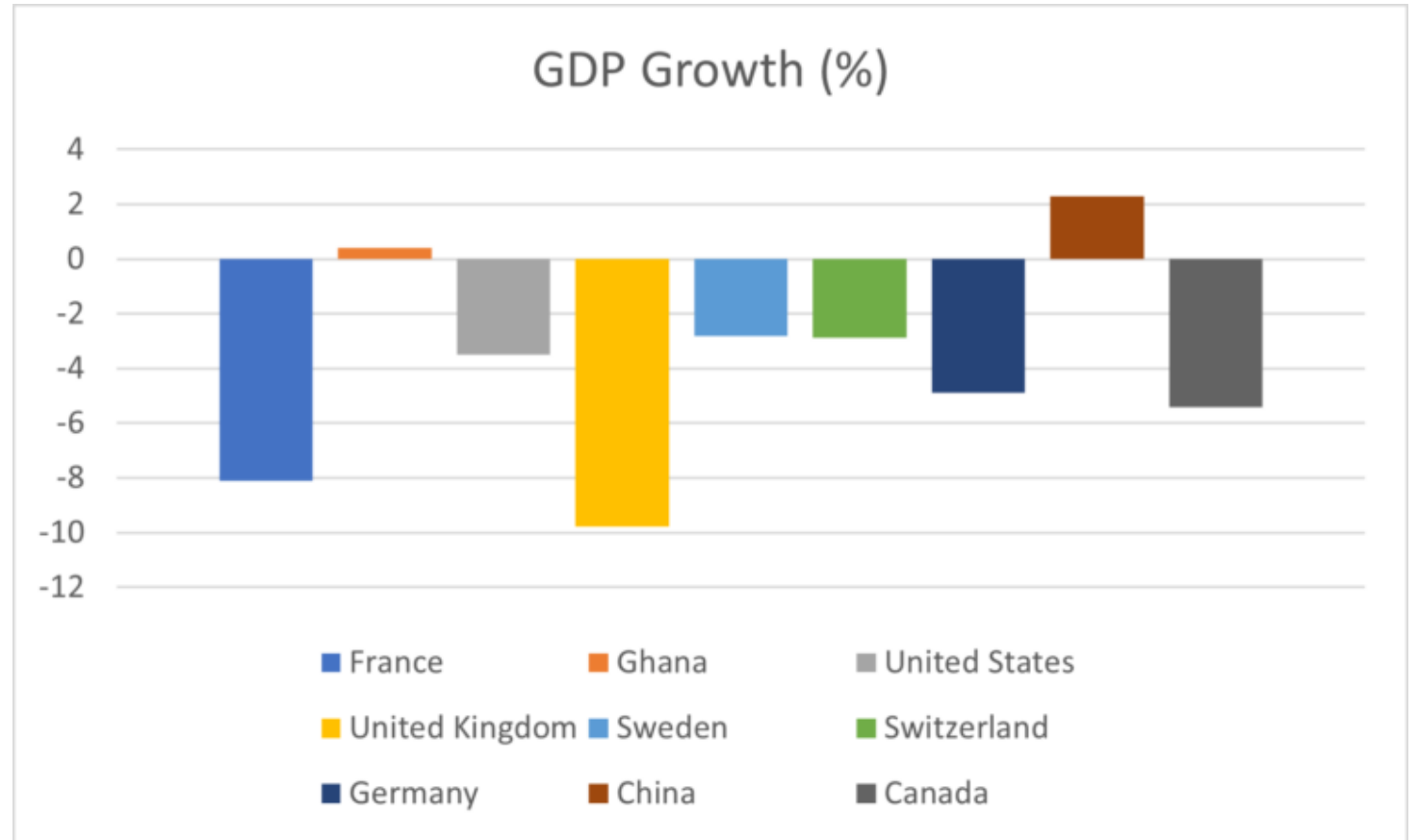
– “I chose the wrong tool”

# Good Chart Choice ?



# Better Alternative

- What can still be improved ?



# The Spaghetti Overload

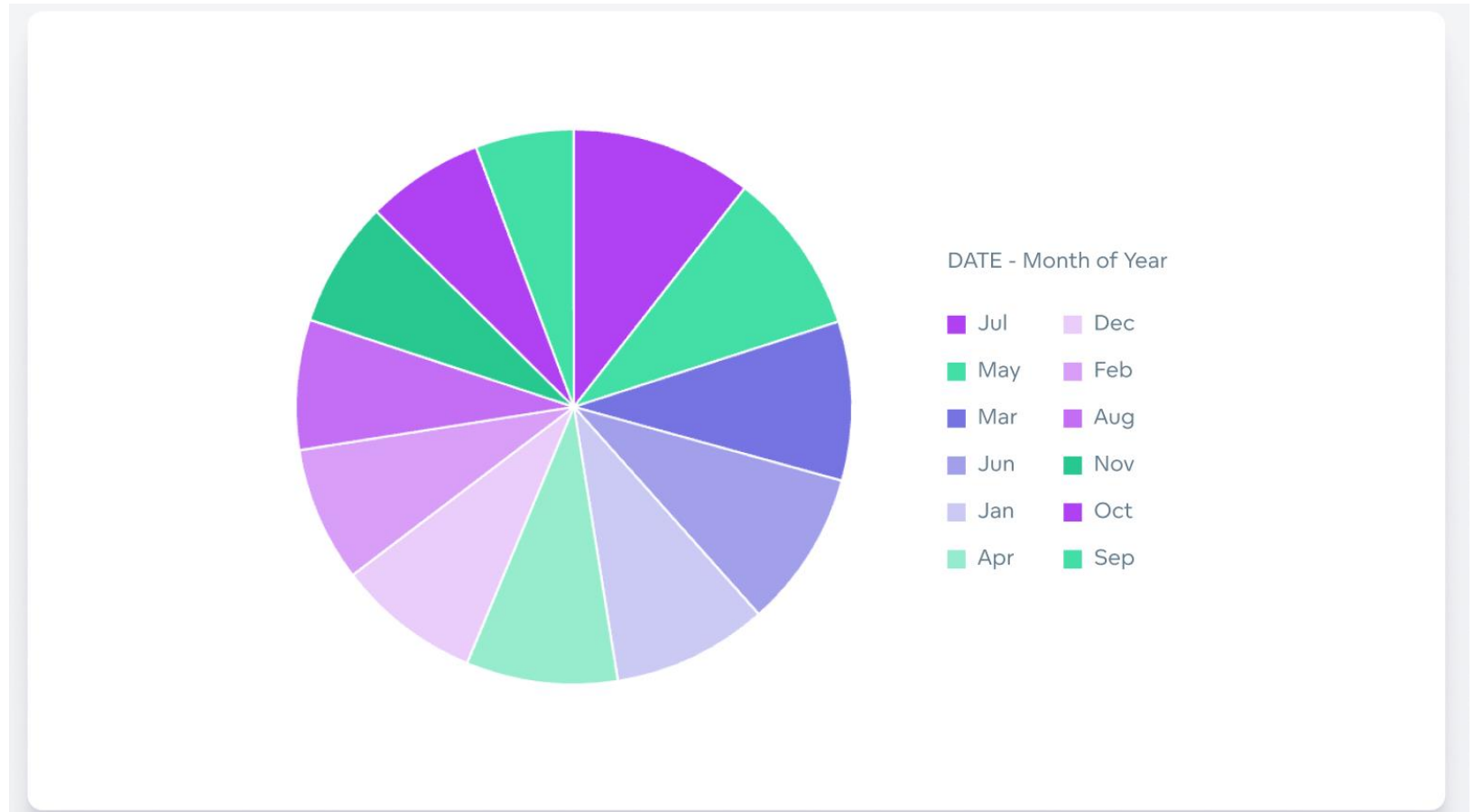
- Poorly chosen visualizations can distort the message of your data
- Impossible to identify any meaningful trends or patterns
- Data must be clear and understandable



# The Infinite Pie

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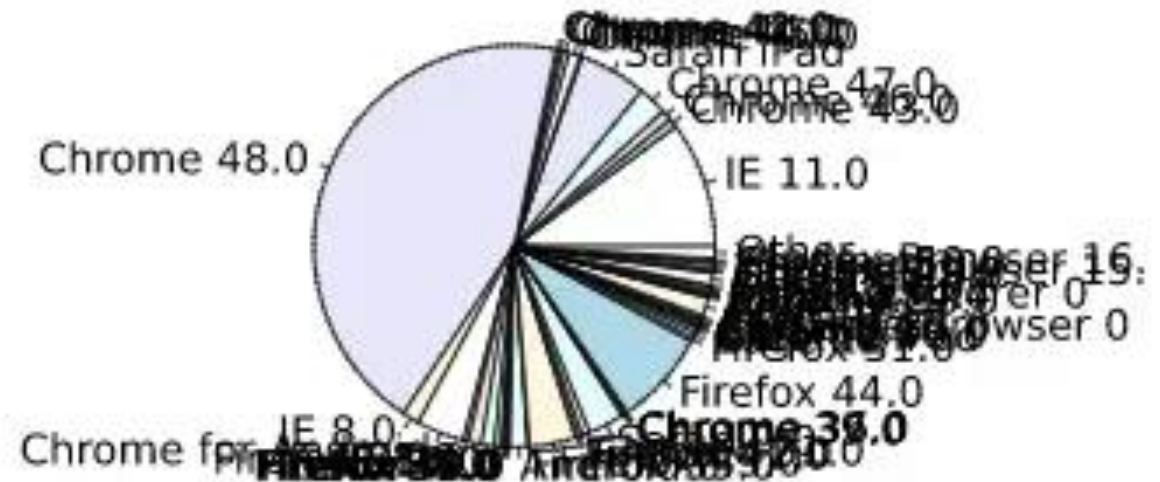
- Too many slices !
- Bad choice of colors !
- Differences are hard to distinguish
- Choose alternative visualizations !!



# Data Suffocation

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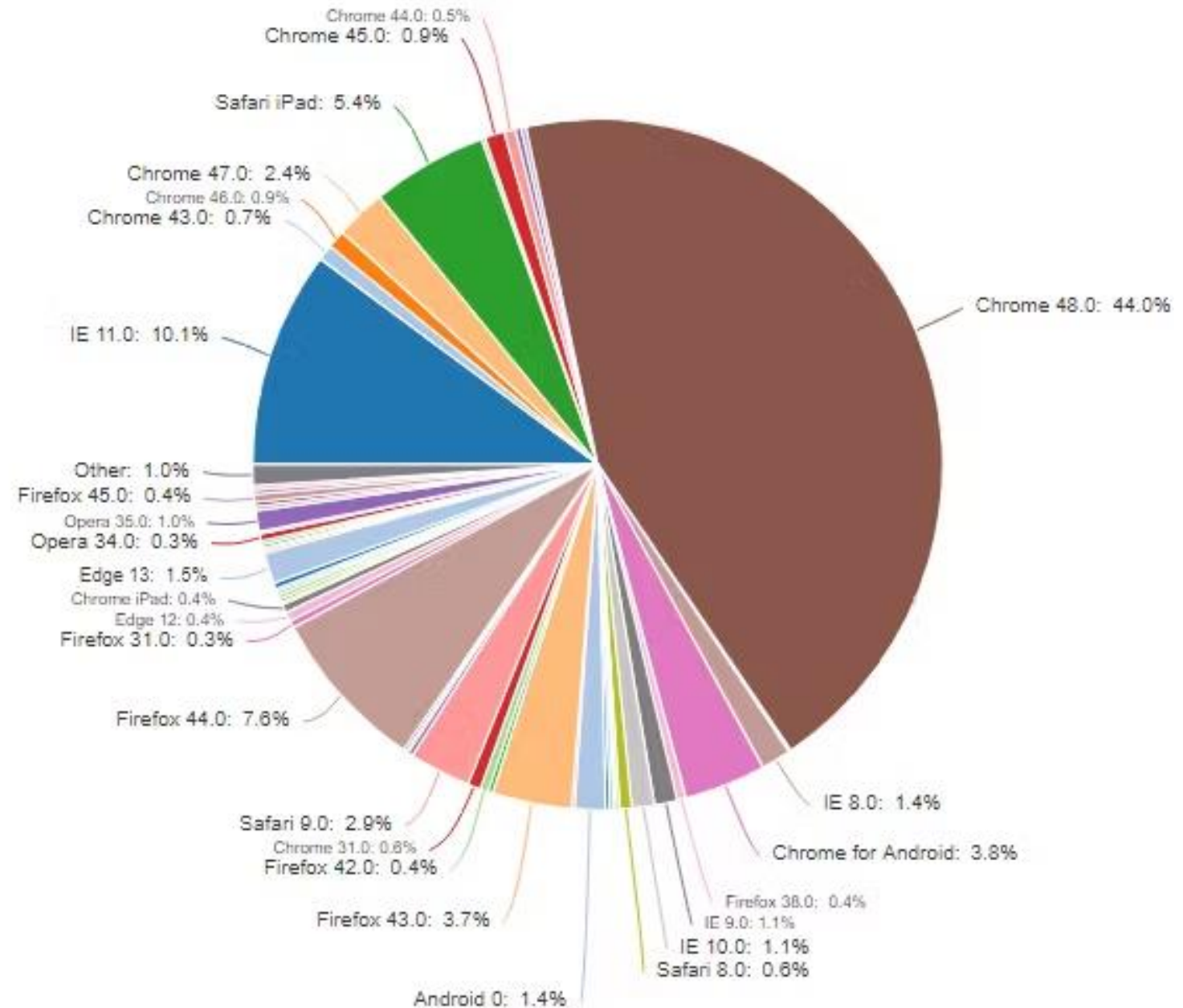
- Data / trends are buried under text, labels, etc.
- Use labels sparingly
- Highlight only the most important data points
- **Data-to-ink-ratio**



# Better ?

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- Changing font size
- Labels connected to data
- No labels for less important data
- **BUT still too many pie sections !!**



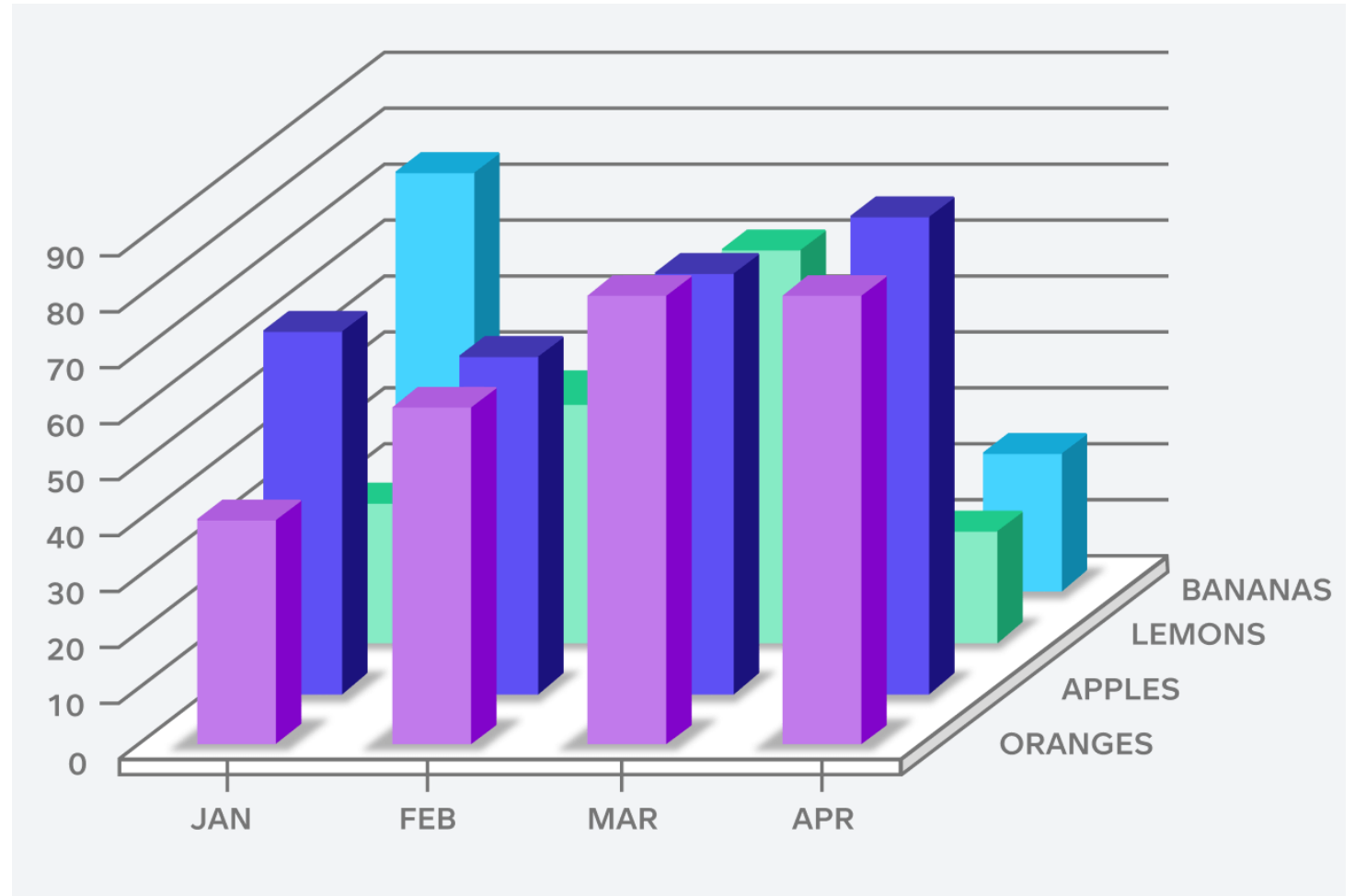


# The Illusion

– “I distorted the physicality”

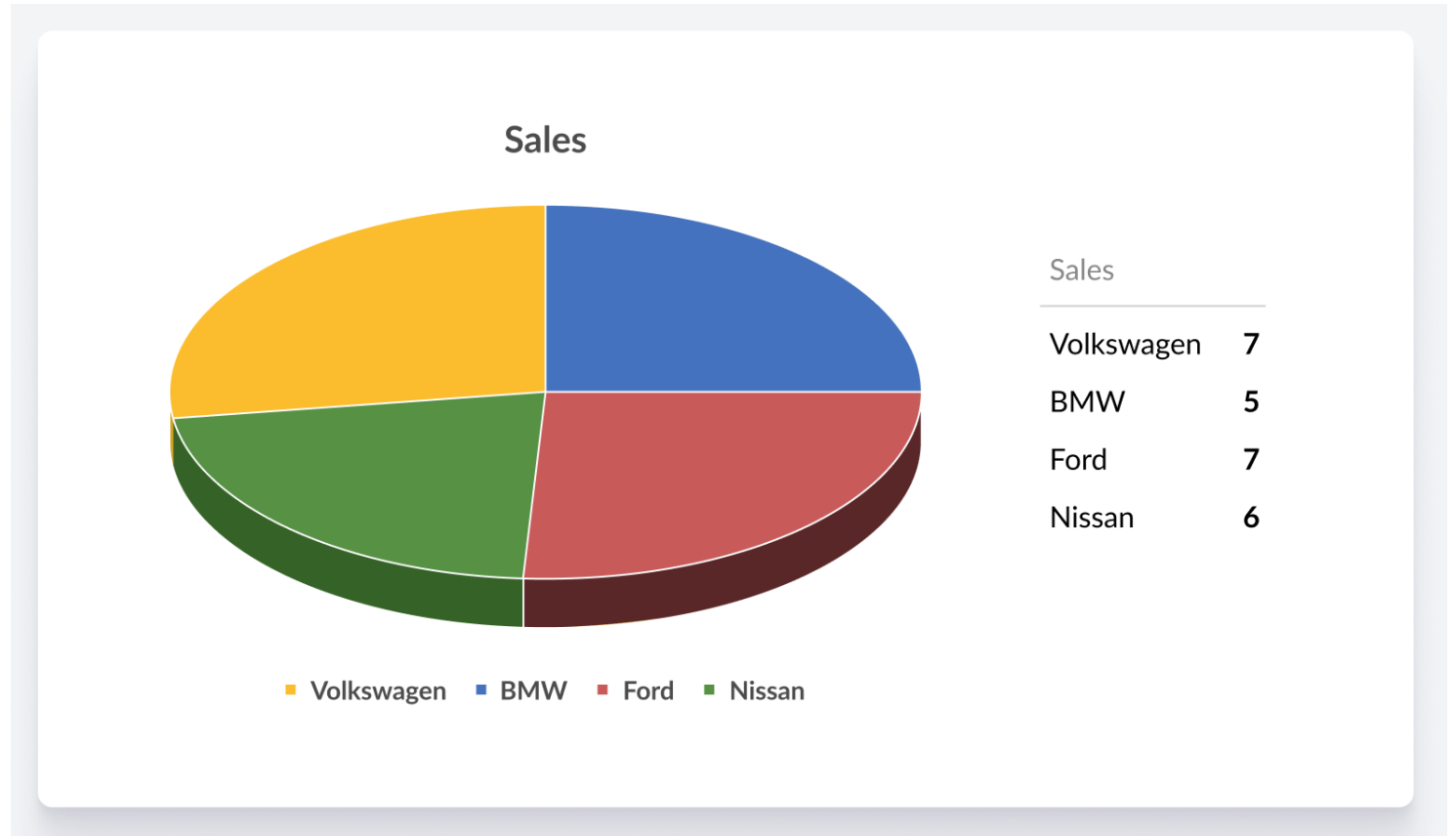
# Perspective Bias

- Perspective distortion
- Cluttered visuals obscure key insights
- The excess of information is overwhelming
- Do not use 3D !
- Use interactive filtering !



# Lying with Perspective

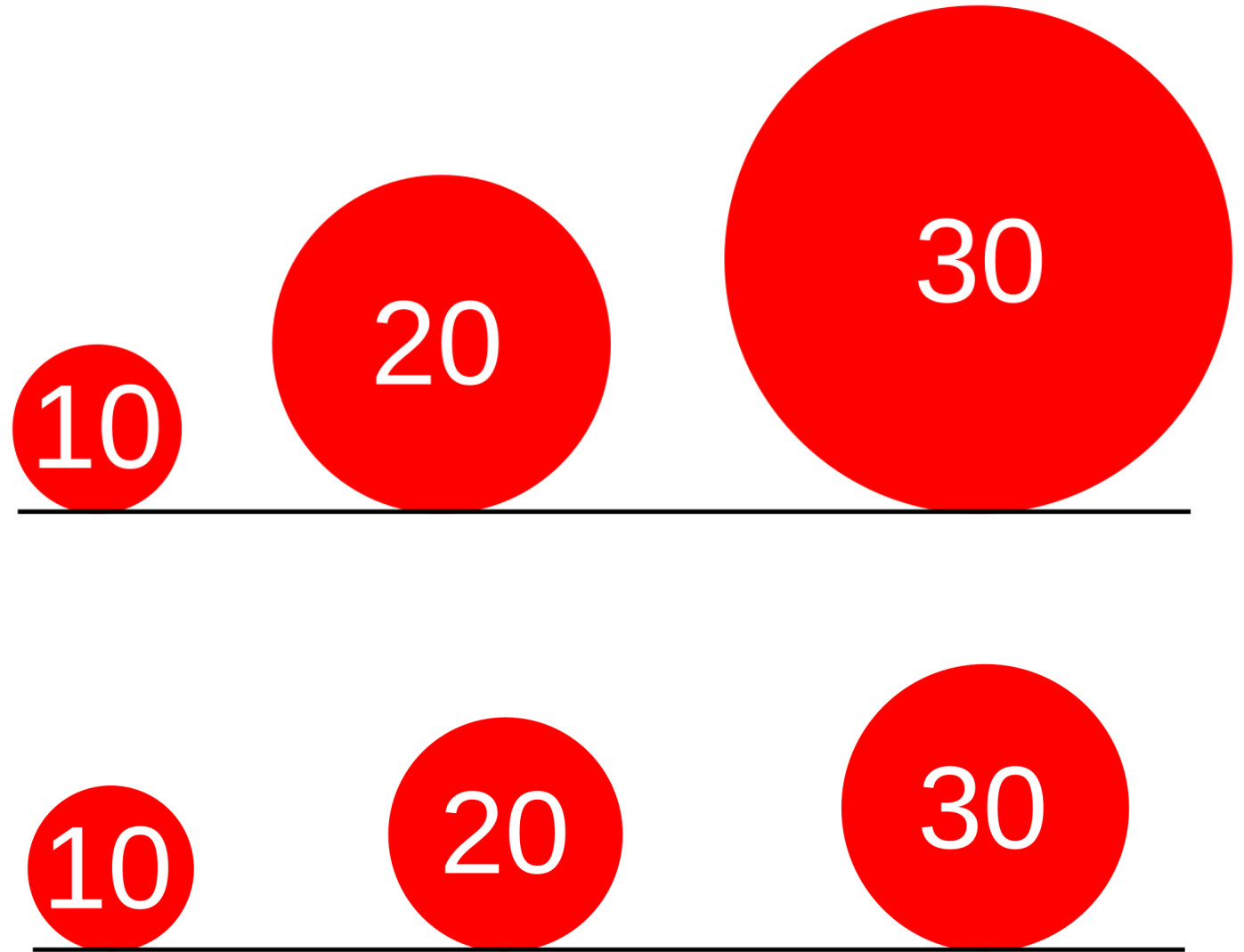
- Perspective distortion leads to wrong interpretation
- Numerical legend is mandatory
- Whenever you're tempted to use 3D, don't !!
- What else can be improved ?



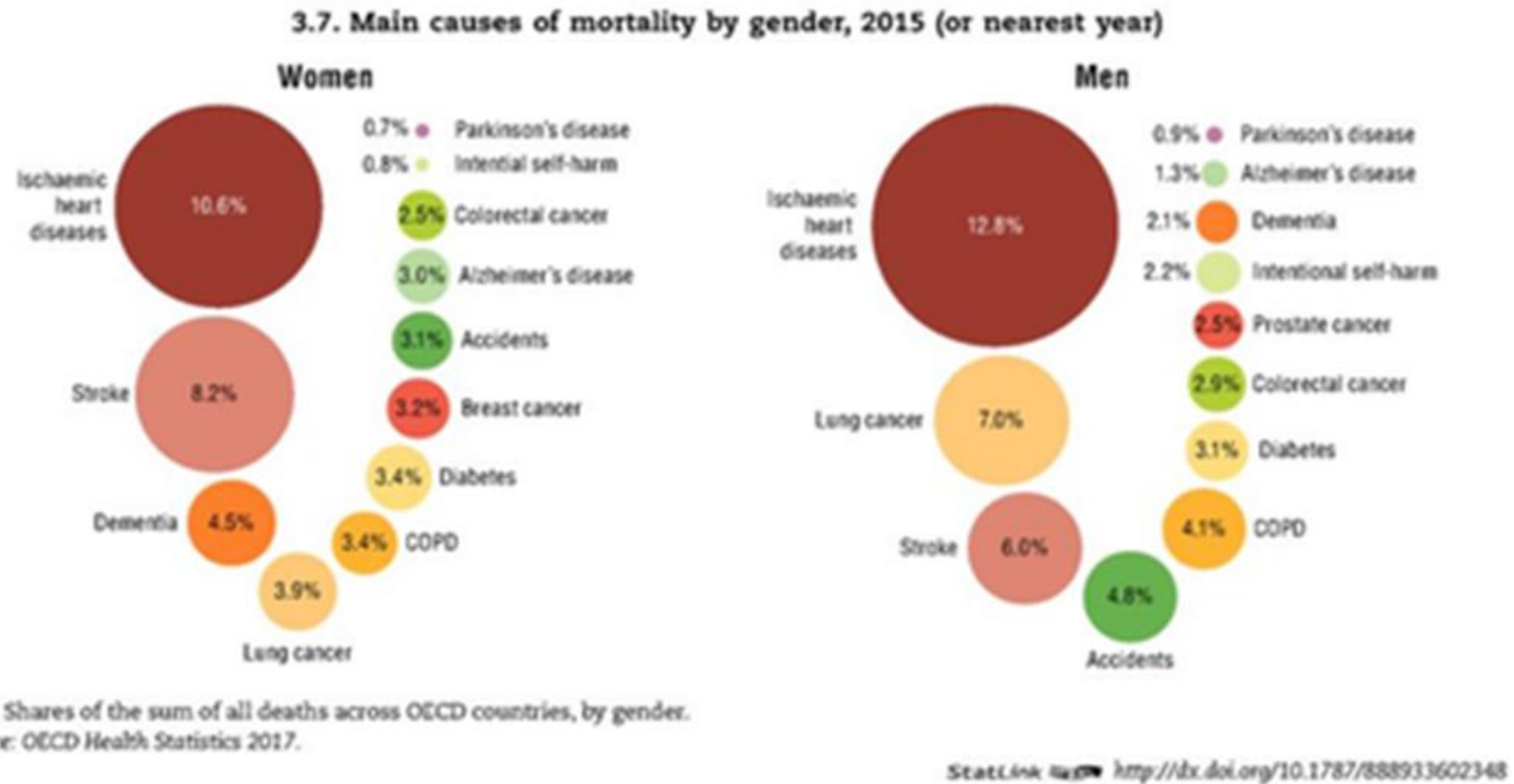
# The Bubble Trap

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- Circles represent data
- **Scale the area of the circle !**
- NOT the radius or diameter
- Humans perceive the **total area**, not just the width !!

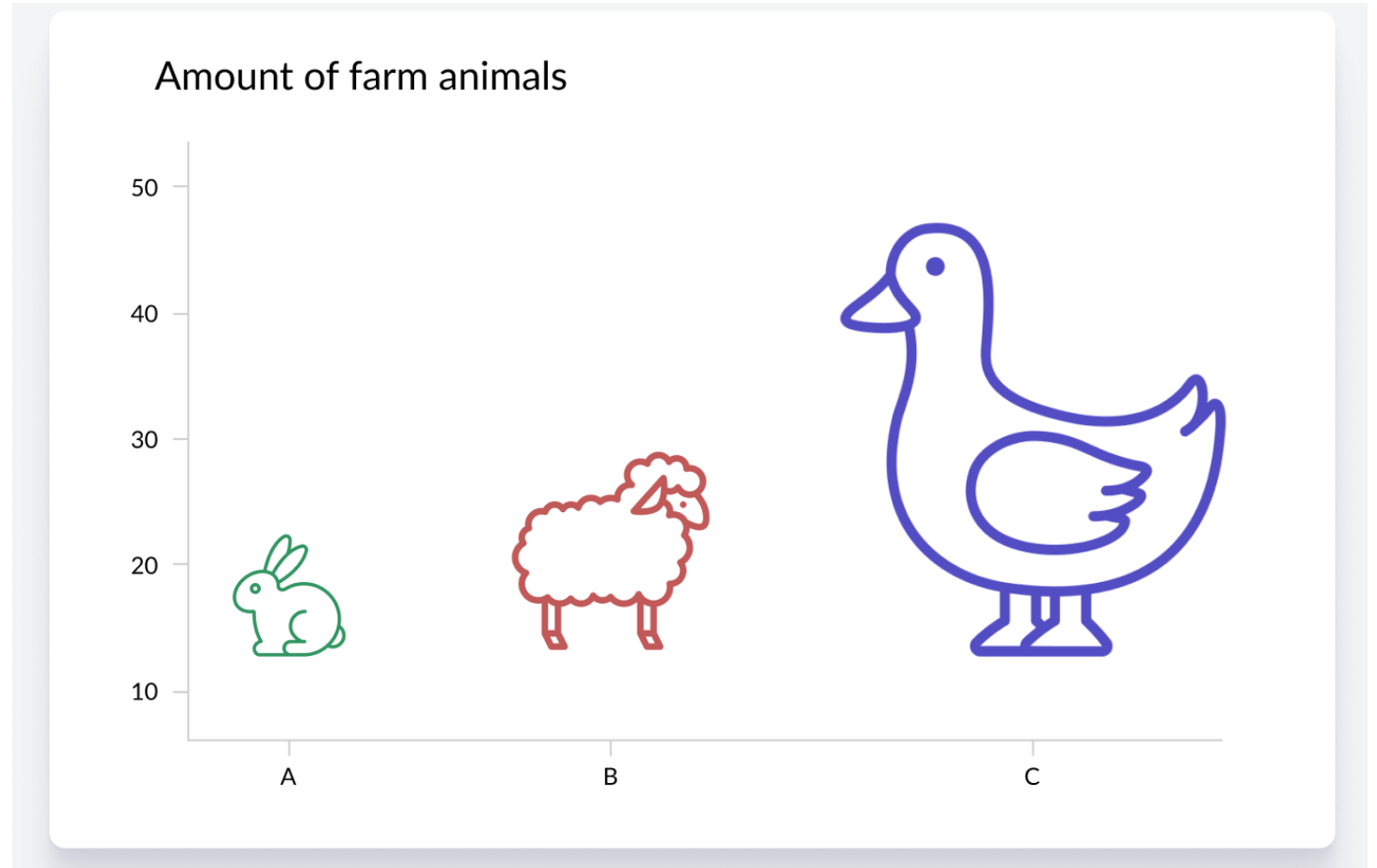


# Official chart, but wrong scaling !



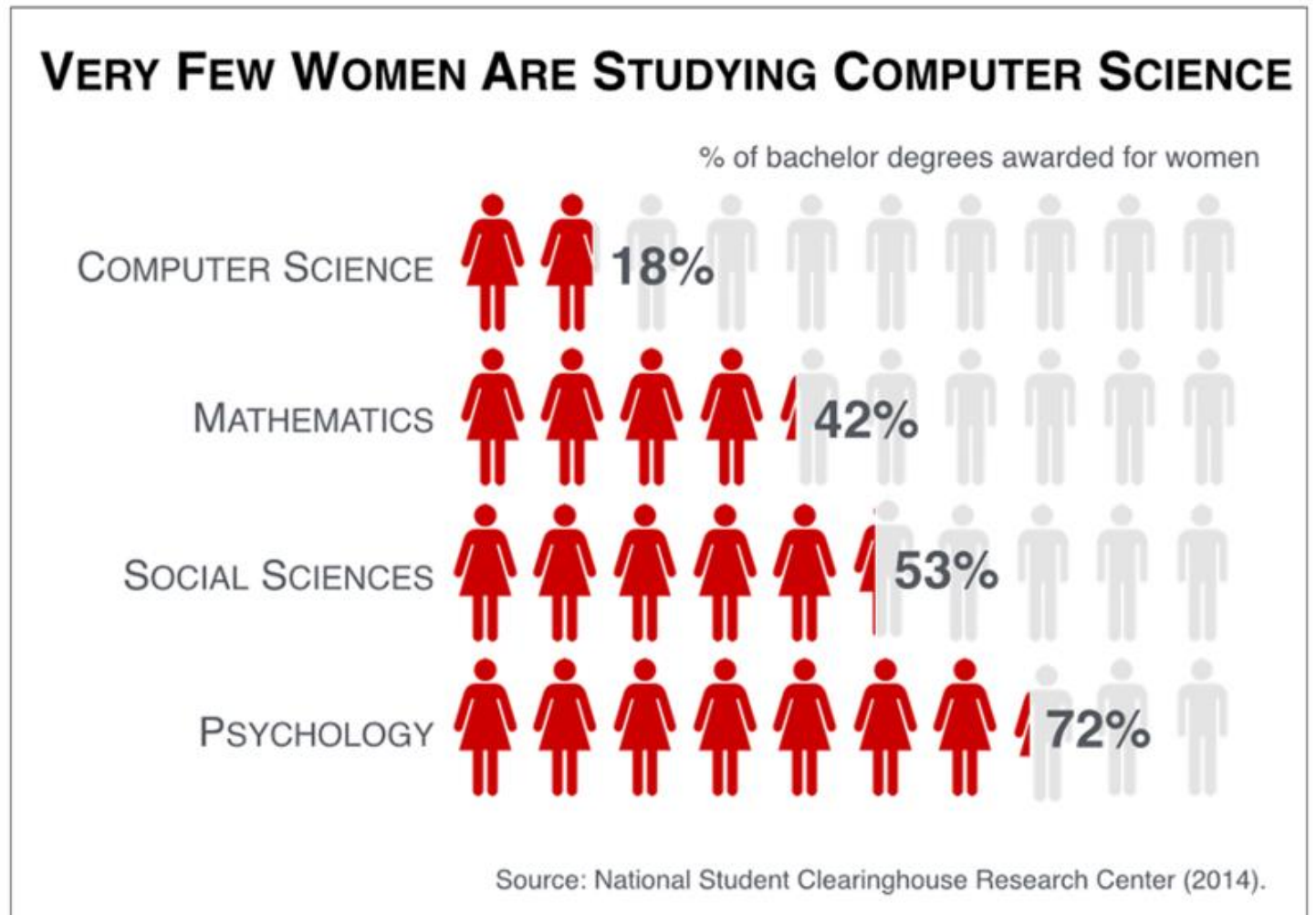
# Misleading Pictograms

- Height vs Area / Width
- Size brings no contribution
- The viewer draws incorrect conclusions !
- Keep the relative scales consistent
- Or use a **unit-based pictogram**



# Unit-Based Pictogram

- Each individual “person” icon has the same size



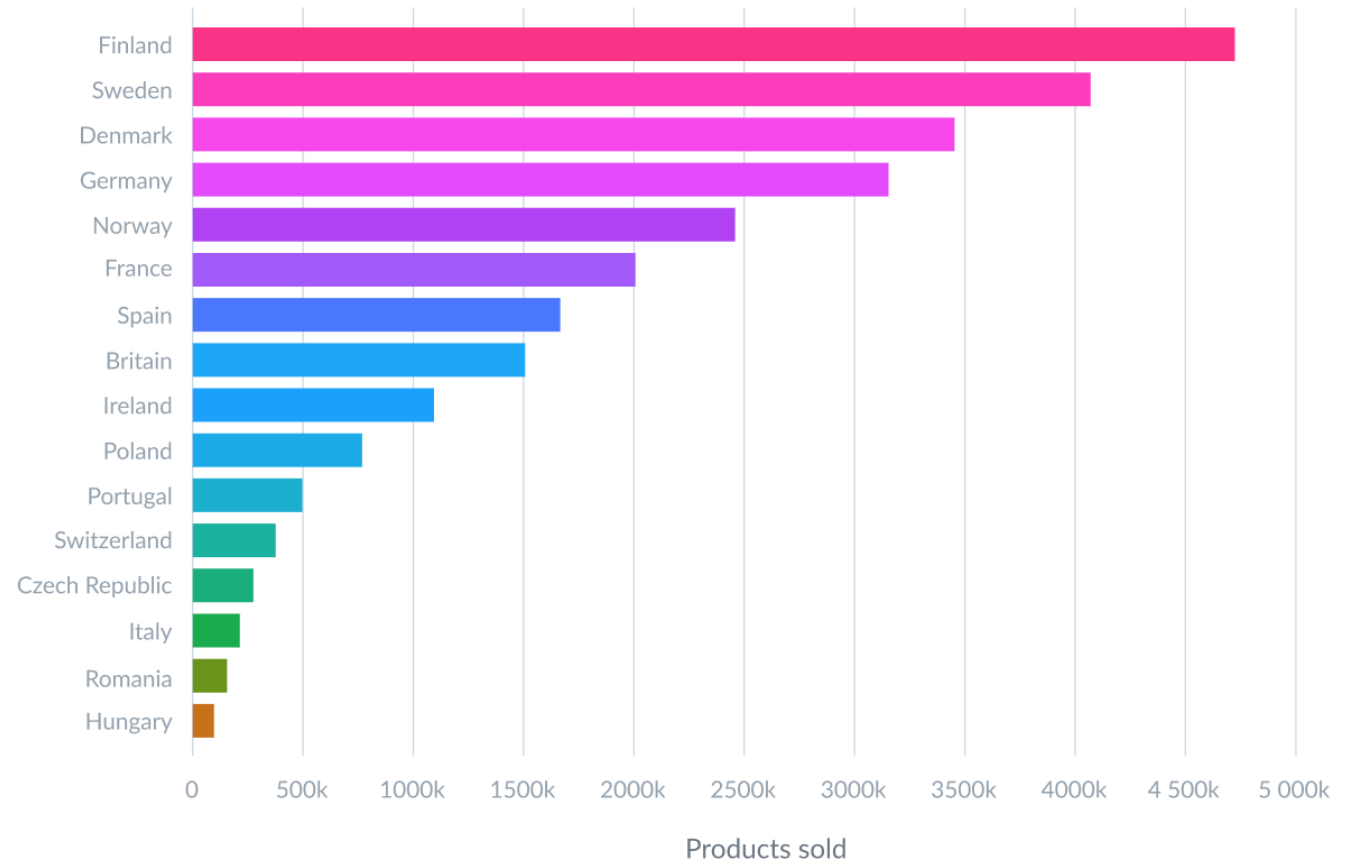
# The Distraction

– “I used color for decoration”

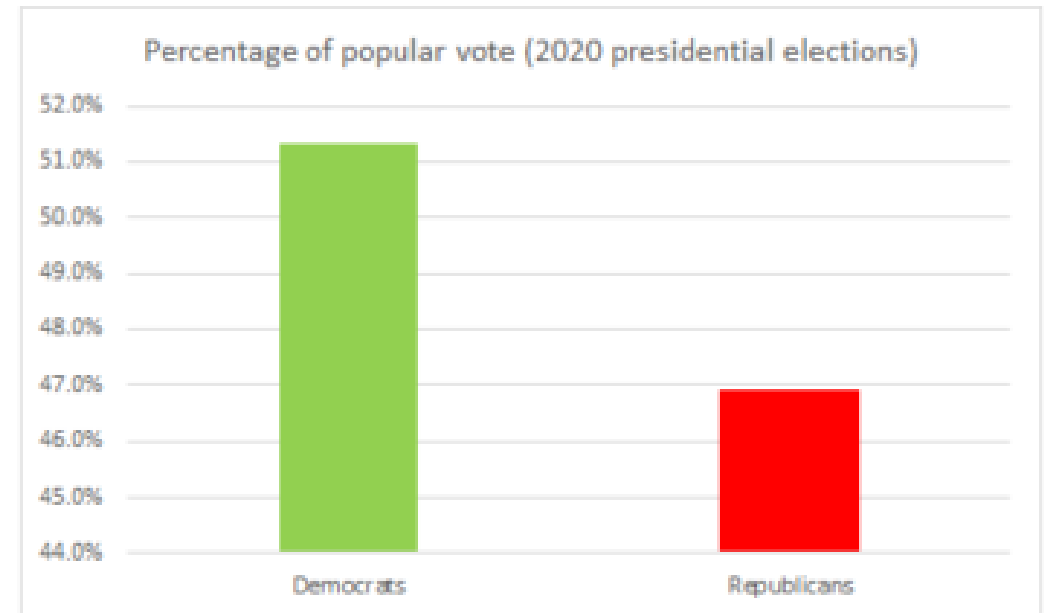
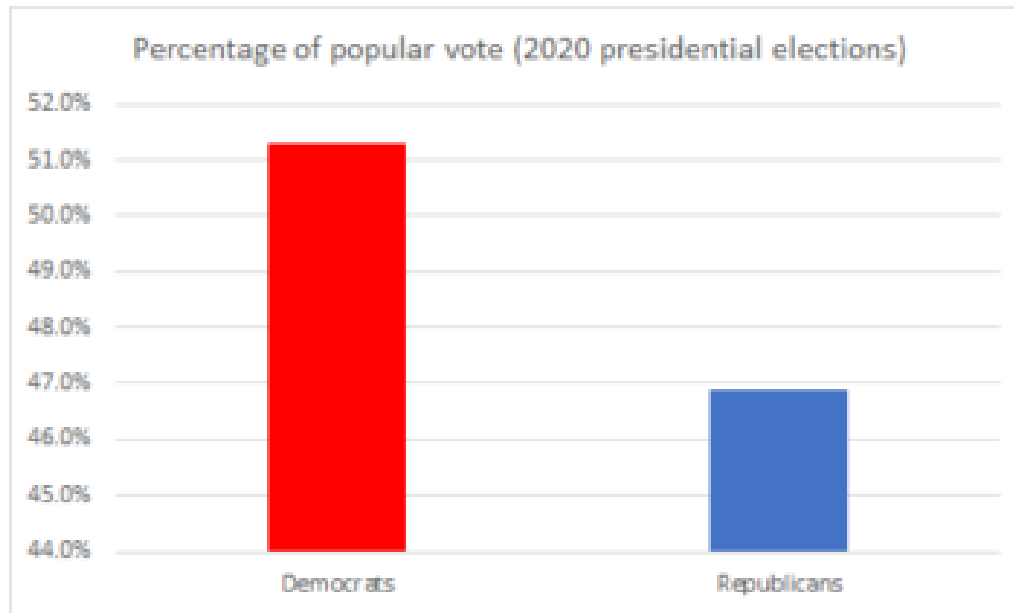


# The Rainbow Chart

- Random color assignment
  - Aesthetically pleasing...
  - NO added clarity !
  - Visual NOISE !
- 
- Color is never a decoration !!
  - Use only to highlight or categorize



# Counter-Intuitive Color Mapping



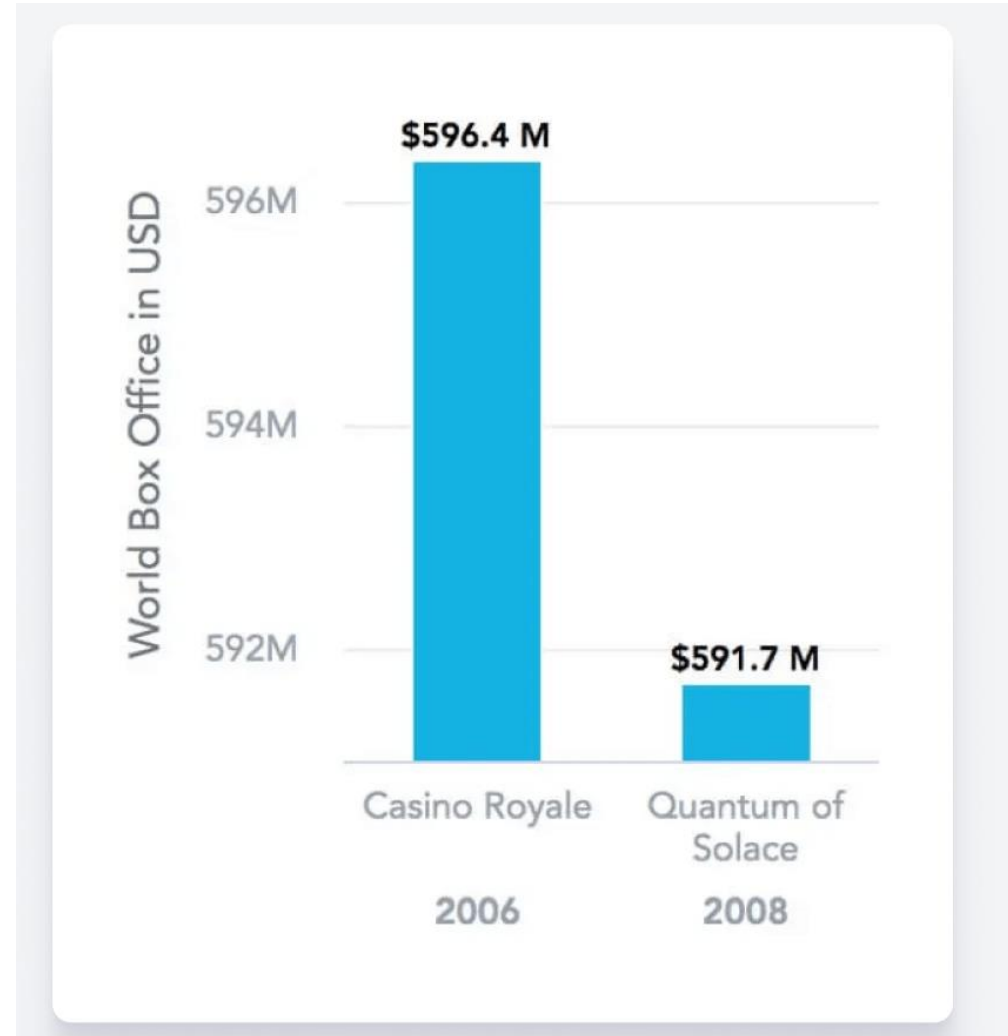
# The Deception

– “I manipulated the Math”

# The Mountain out of a Molehill

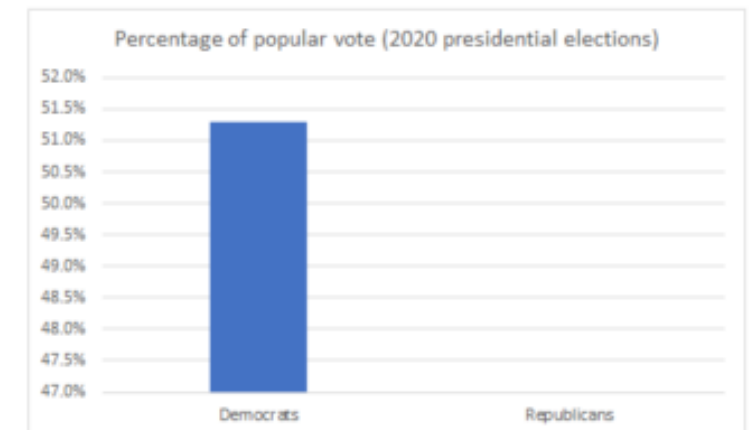
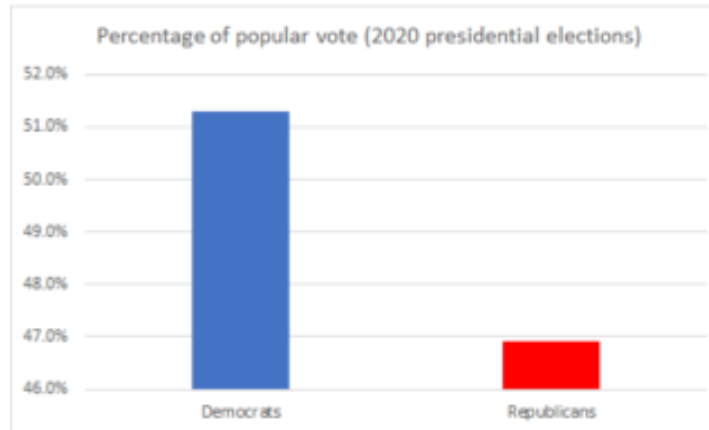
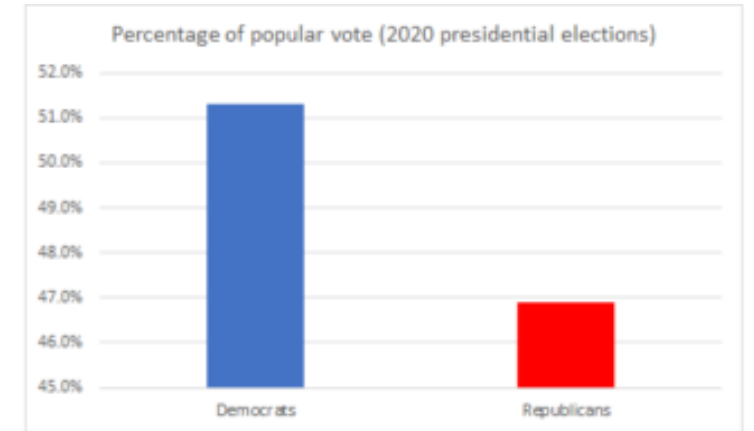
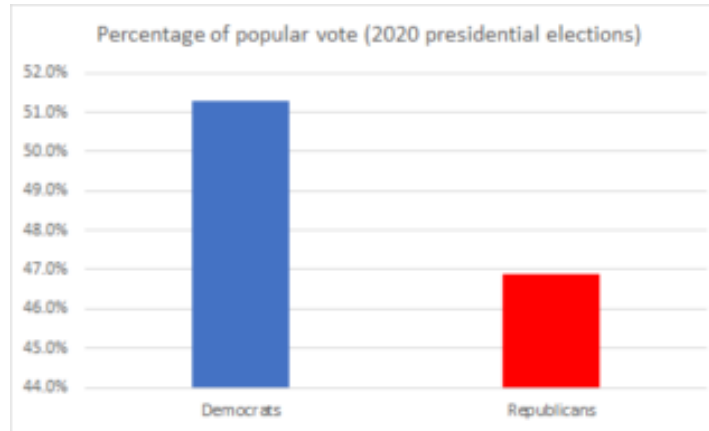
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- Difference is exaggerated
- Disparity appears greater than it is
- Leading to an inaccurate conclusion
- This is sometimes done intentionally !

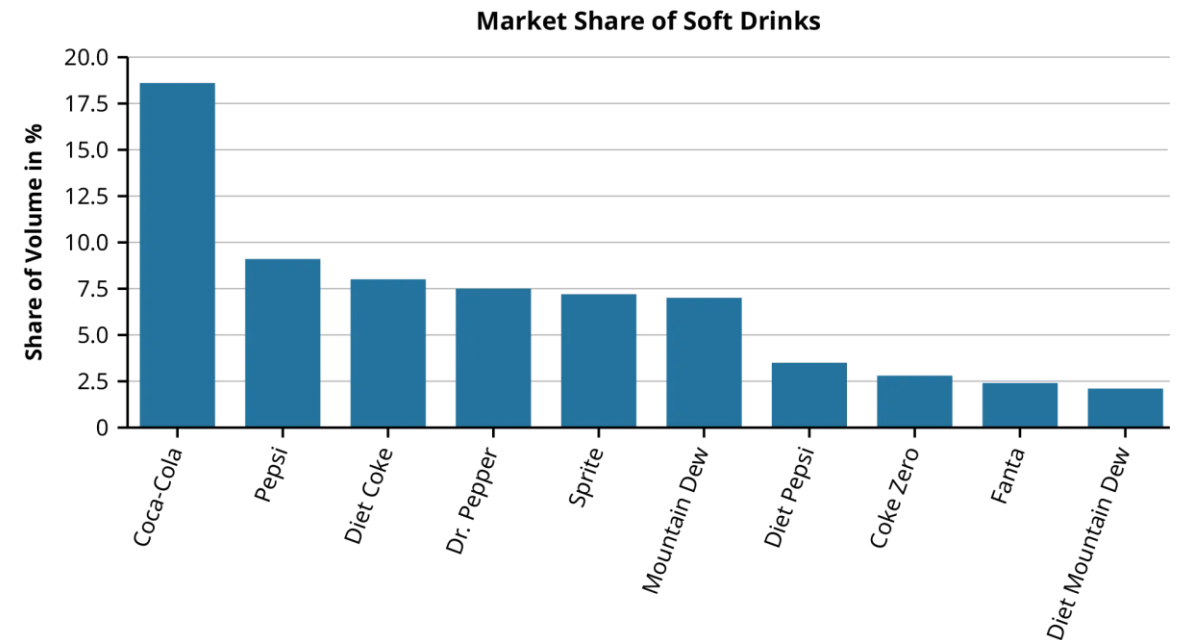
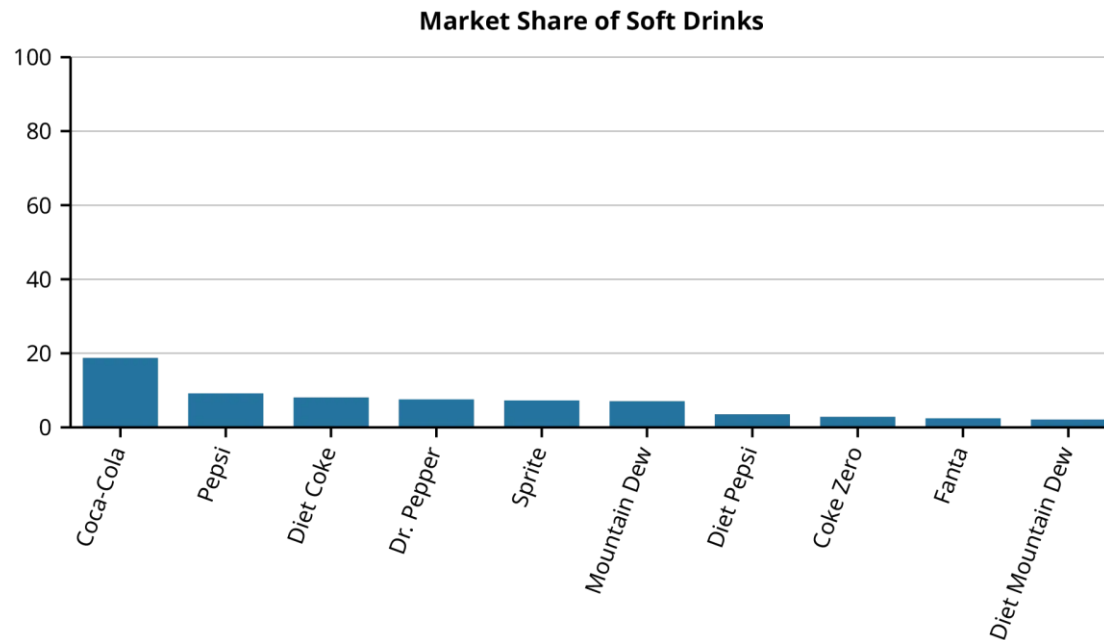


# What is the percentual difference ?

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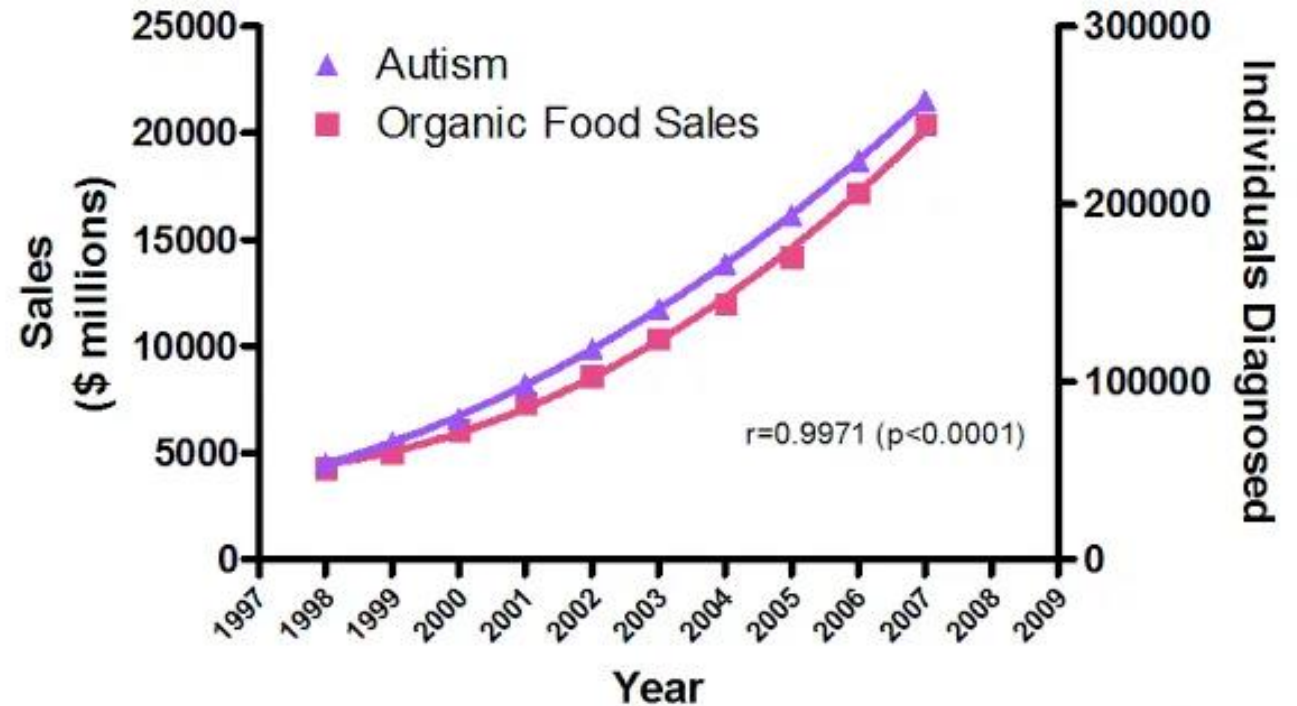


# Hiding Differences through Vertical Axis Scaling



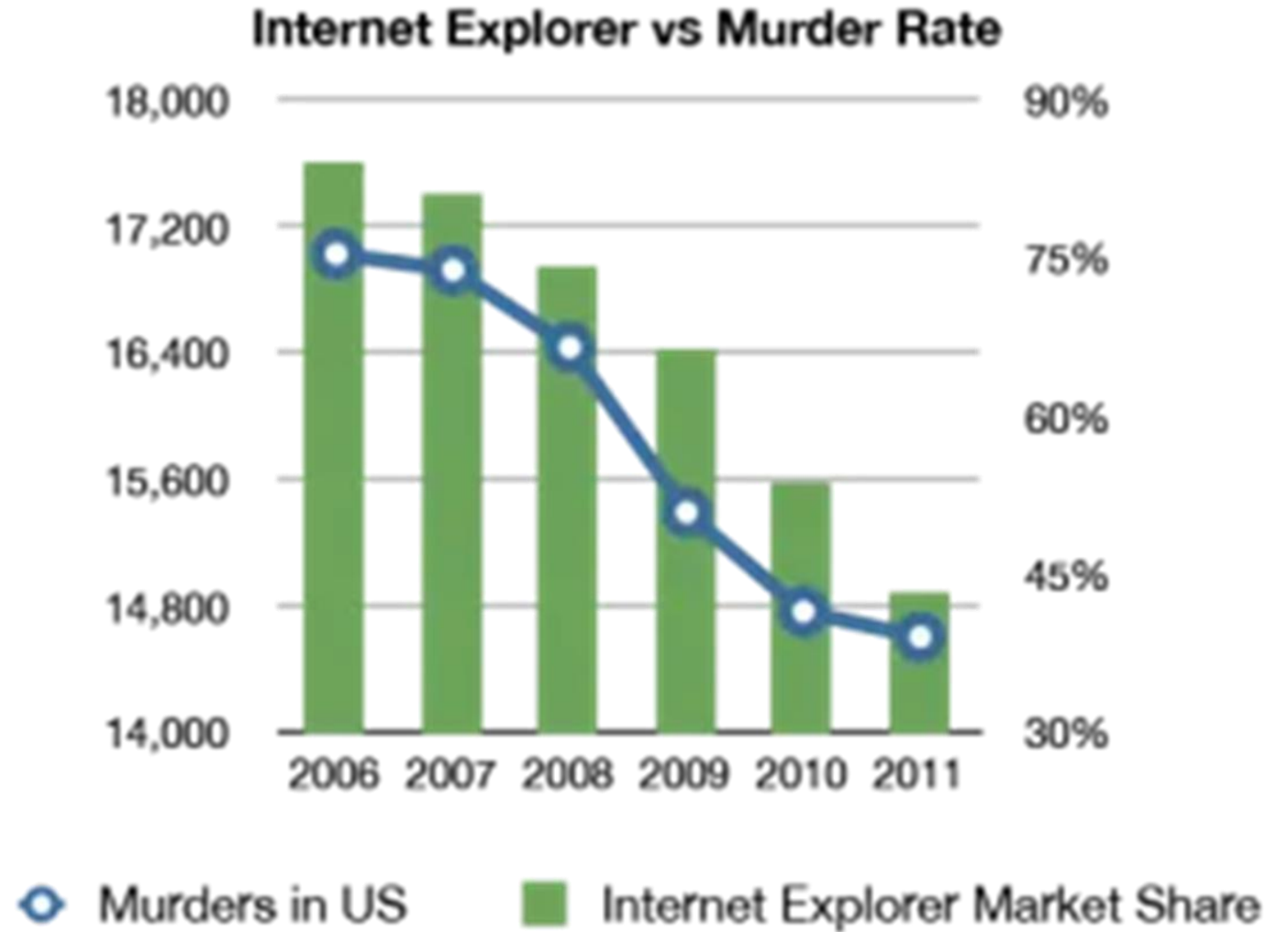
Do organic foods cause autism ?

The real cause of increasing autism prevalence?



Sources: Organic Trade Association, 2011 Organic Industry Survey; U.S. Department of Education, Office of Special Education Programs, Data Analysis System (DANS), OMB# 1820-0043; \*Children with Disabilities Receiving Special Education Under Part B of the Individuals with Disabilities Education Act

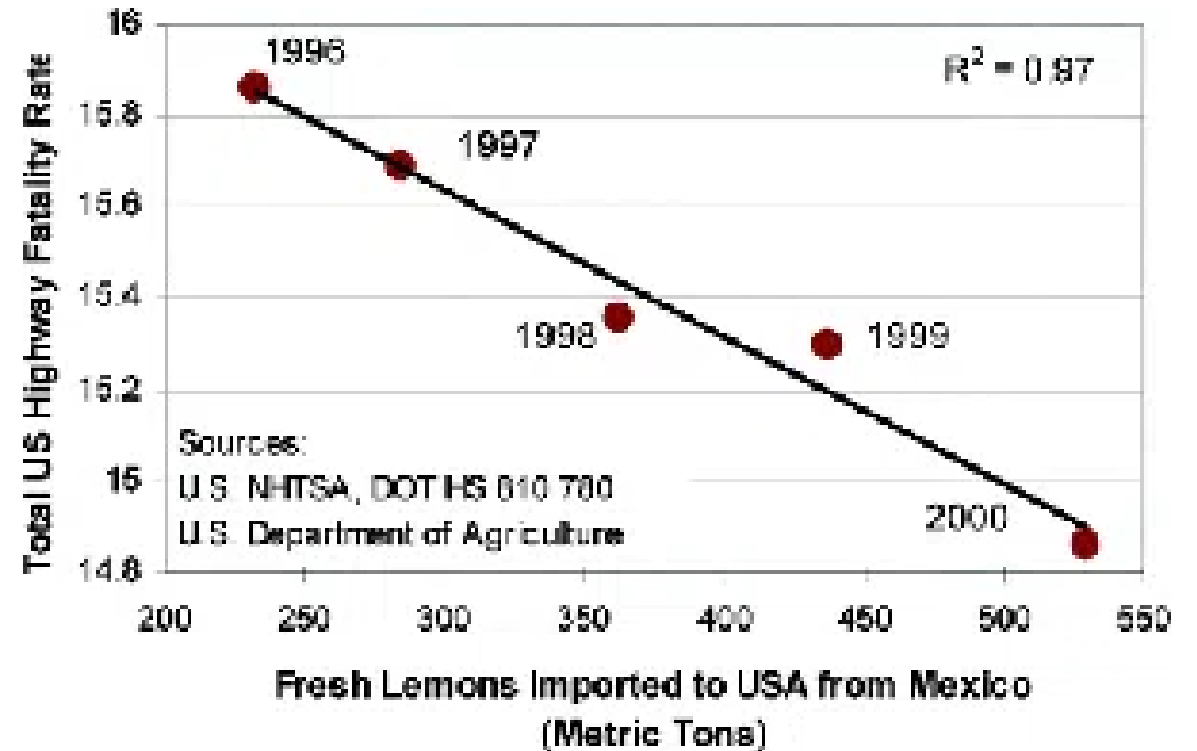
Is Murder Rate  
connected to  
Browser Use ?





# Axis Distortion

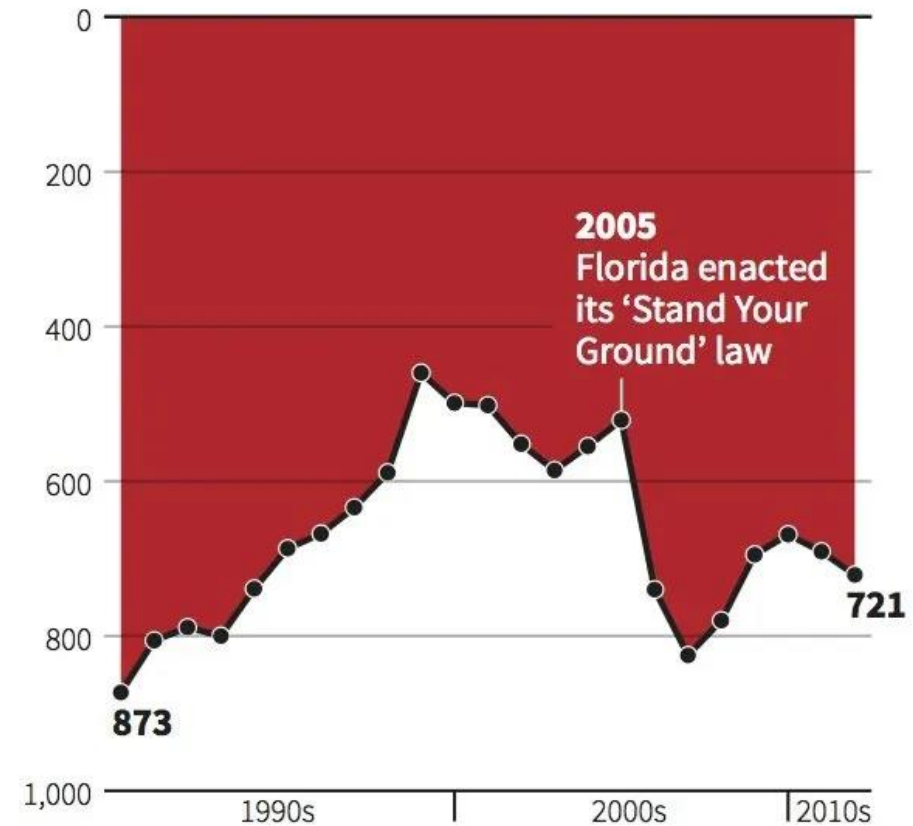
- Stretching / compressing different scales
- Force a correlation that does not exist !!
- **Correlation is not causation !**



Did gun deaths go up or down after 2005 ?

## Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

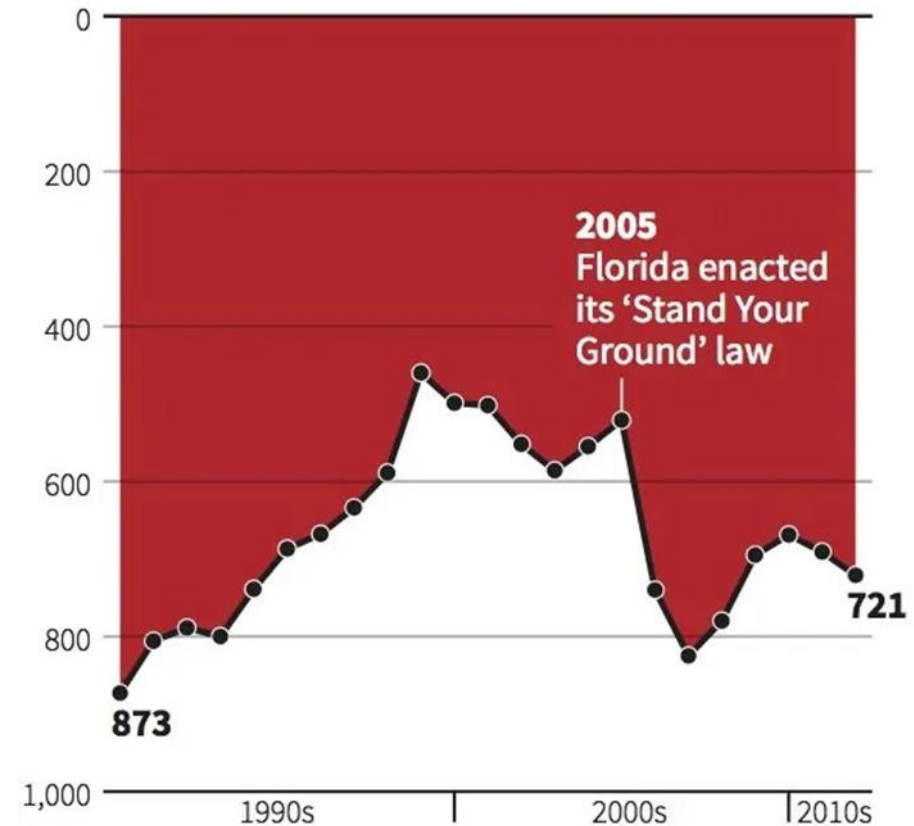
REUTERS

# The Upside-Down Truth

- Zero is at the top !!

## Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

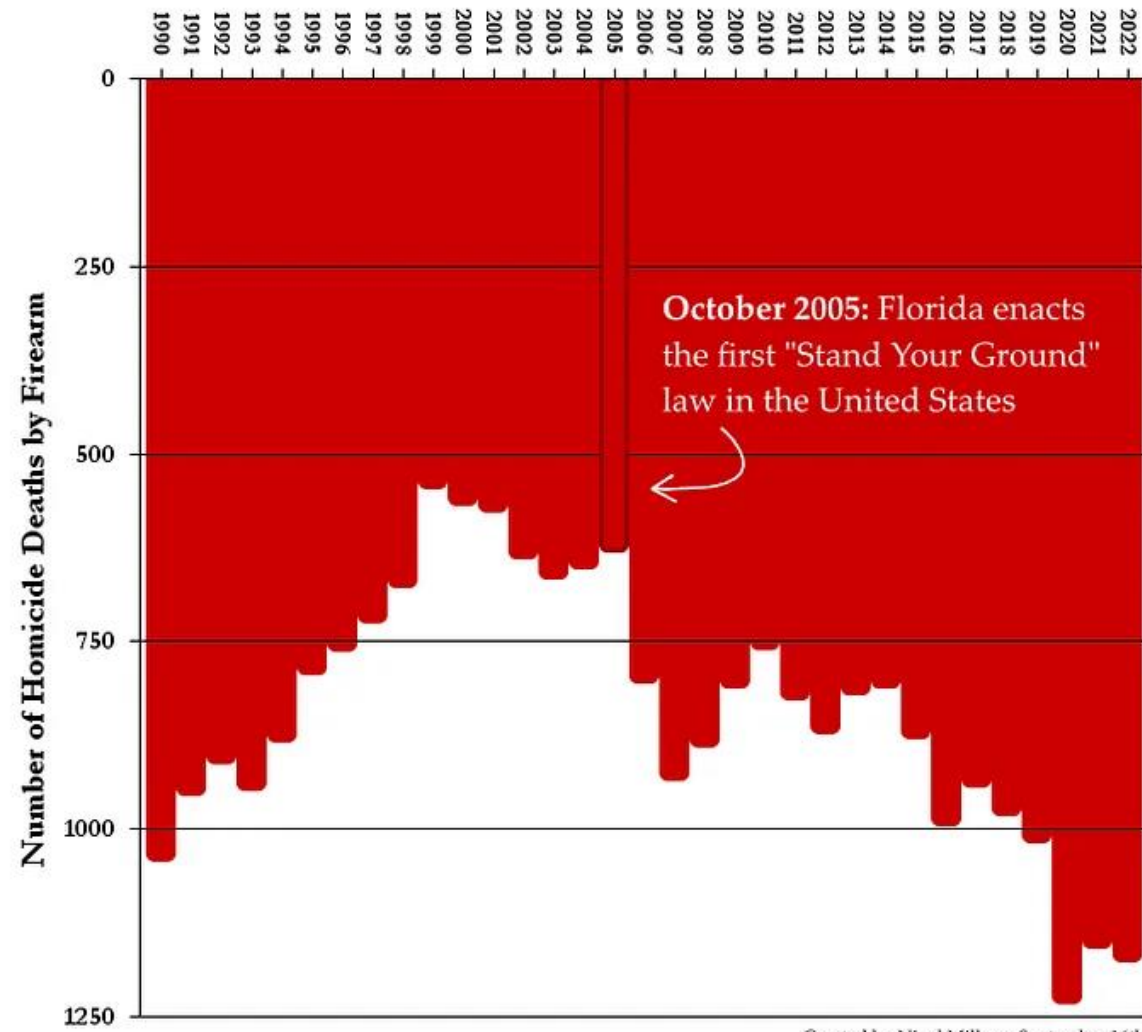
REUTERS

# Better ?

- Why ?

## Gun Deaths in Florida Before vs. After "Stand Your Ground"

1990 - 2022, Raw Count



Created by Nigel Mills on September 16th, 2024.

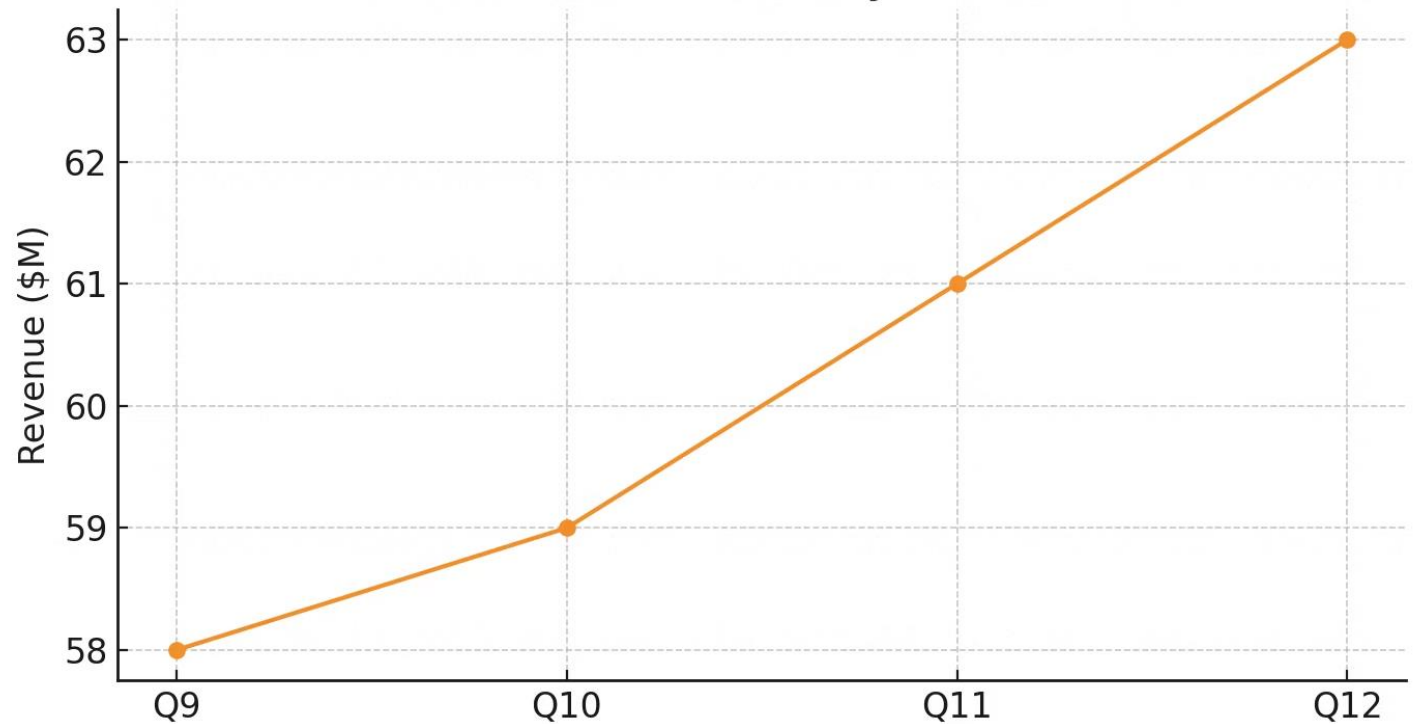
Based on graphs by Christine Chan and Simon Scarr.

Data sourced from USAFacts, FLHealthCharts, and the Florida Department of Health, Bureau of Vital Statistics.

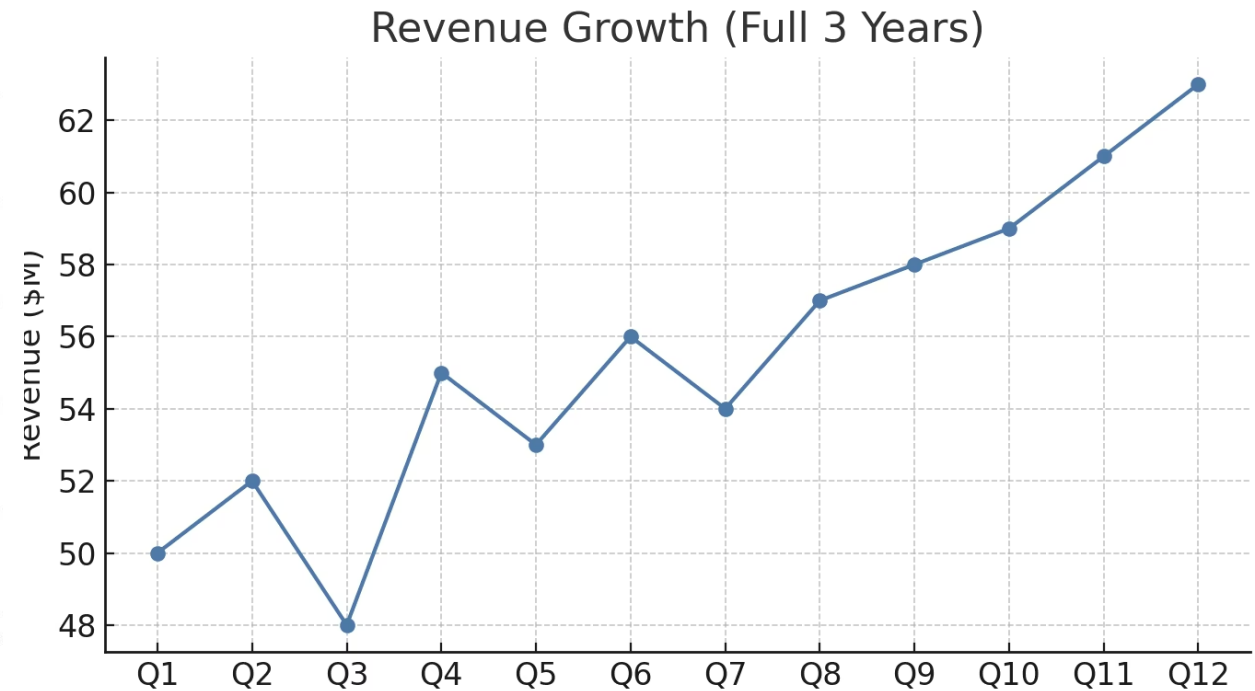
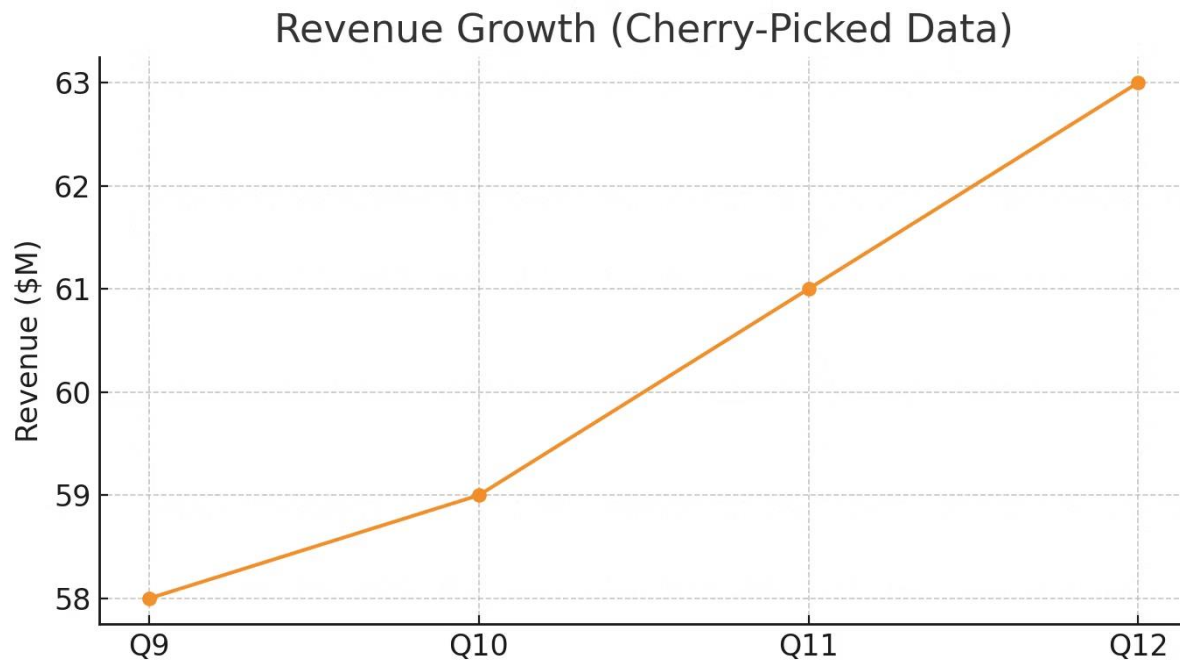
# Cherry-Picking

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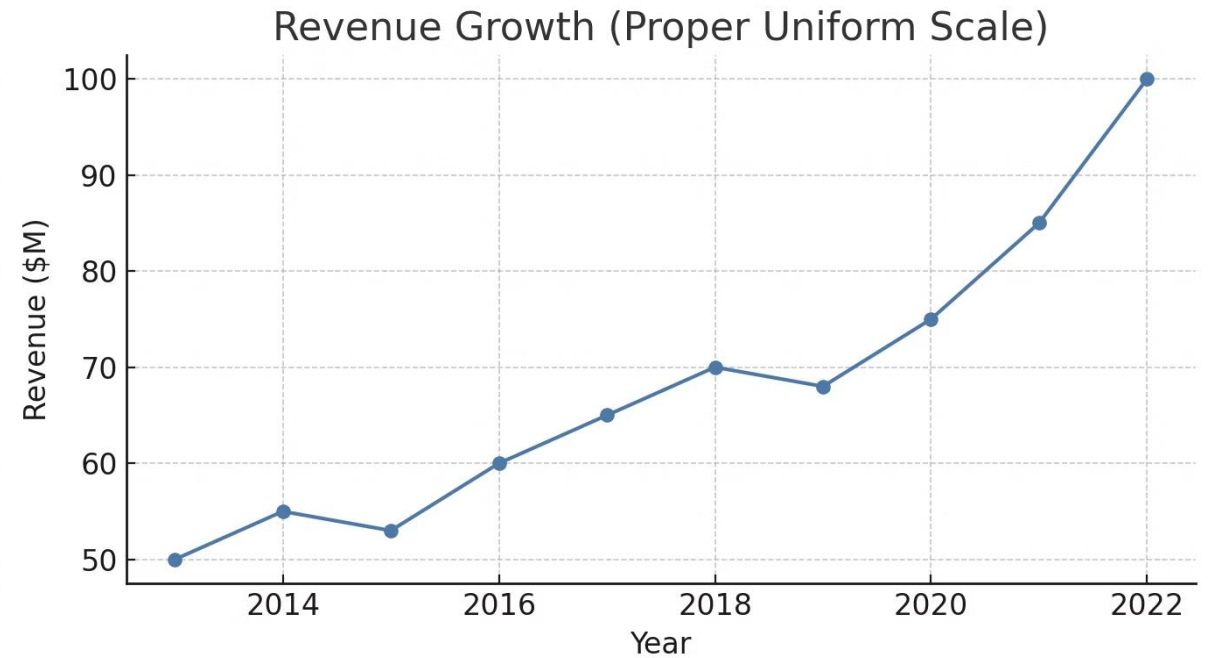
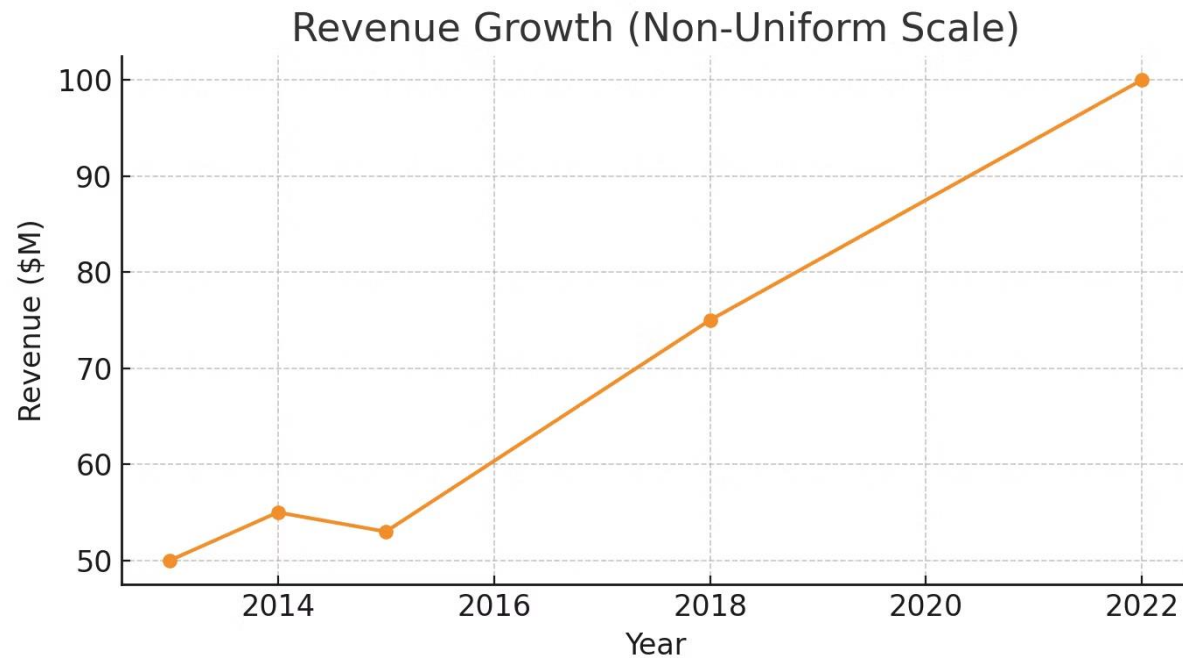
- Selected data range ?
- Length of time window ?
- Volatile vs stable trends
- Temporary dips vs crashes
- **Context matters !!**
- What happened before / after this timeframe ?



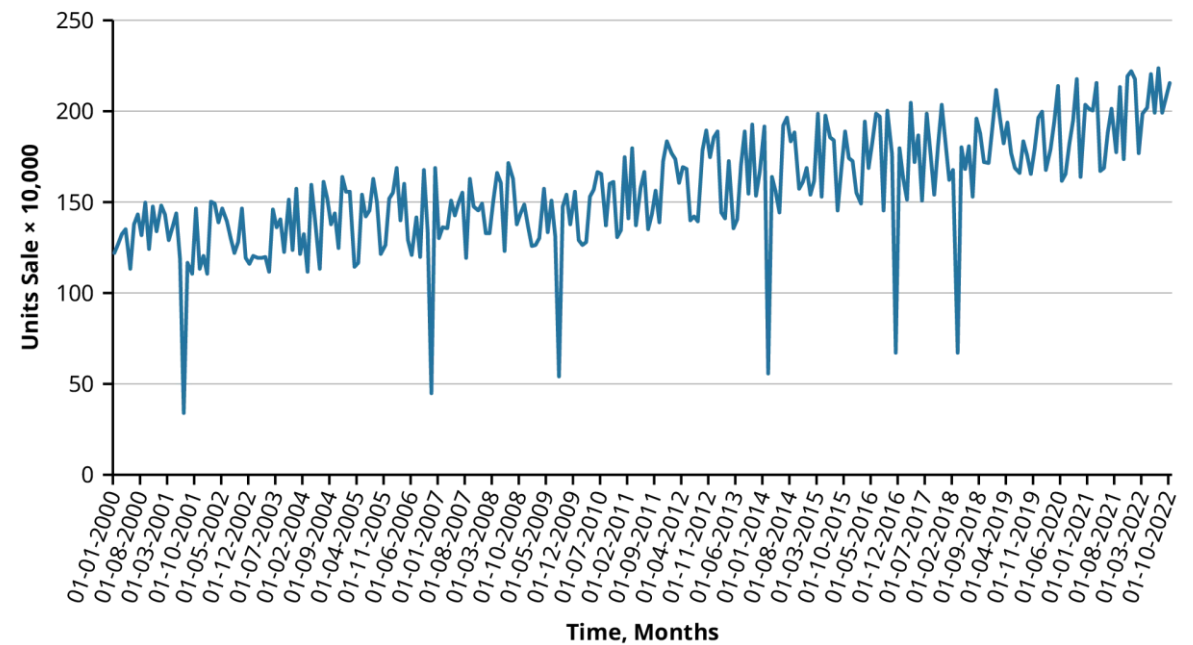
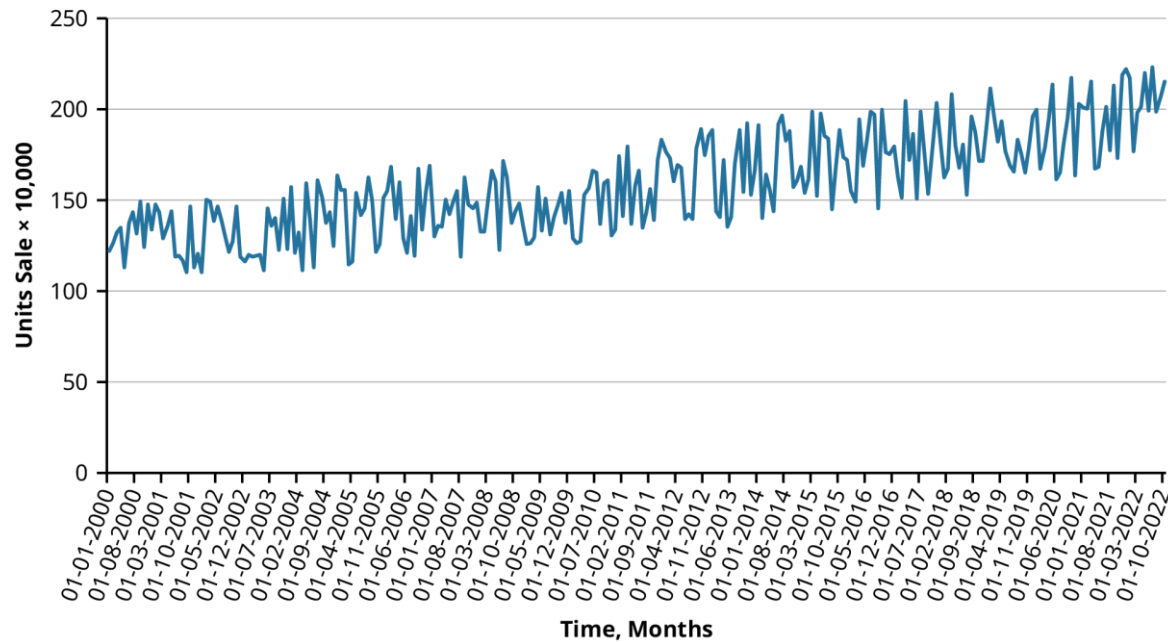
# Disclosing the Truth



# Cherry-Picking – Non-Uniform vs Uniform Spacing



# Cherry-Picking – Removing Outliers



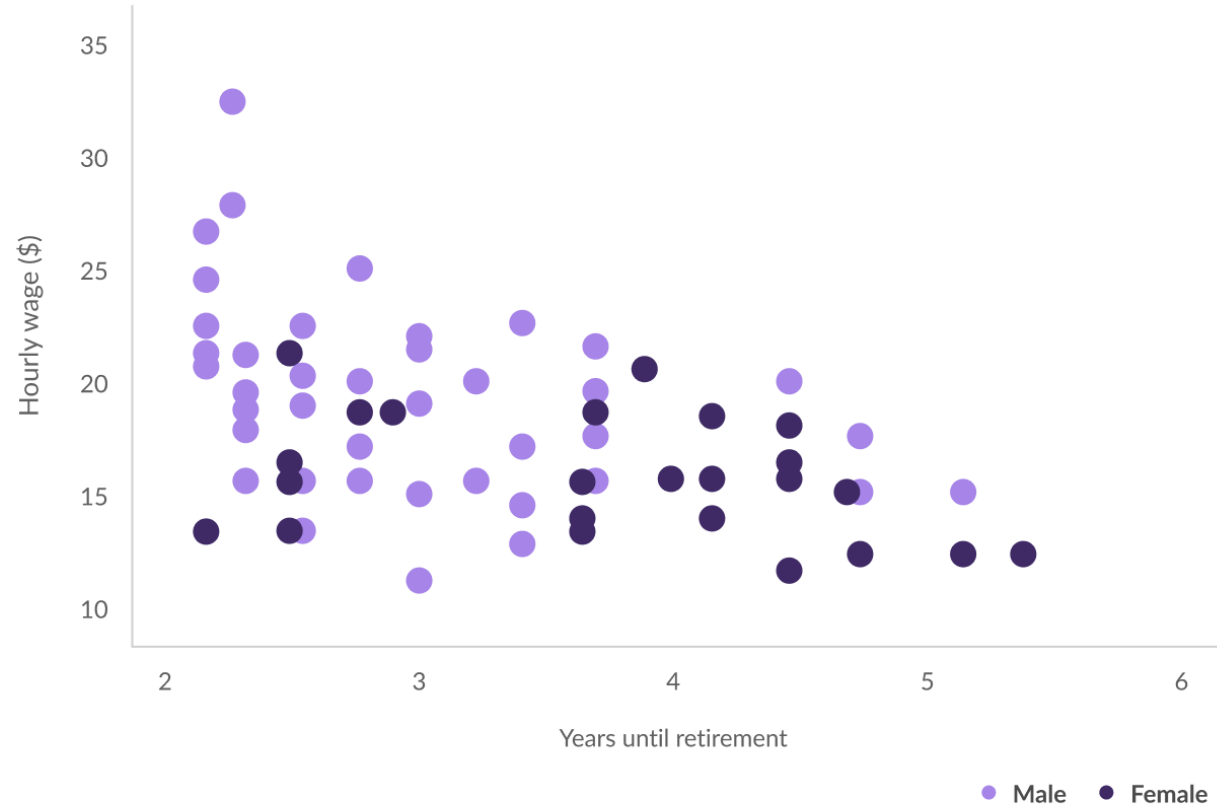


# The Confusion

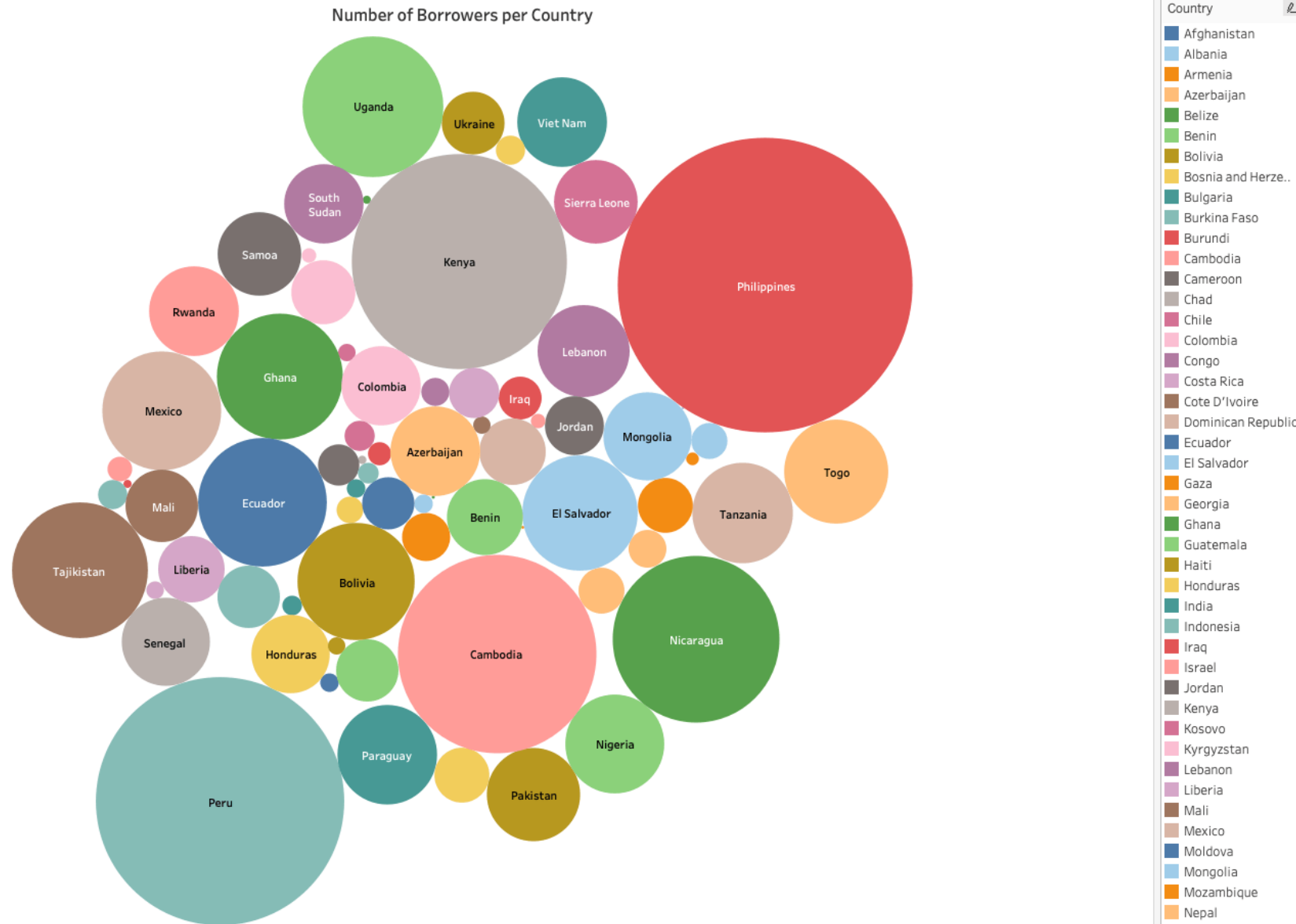
– “I did not make things clear”

# Overlapping Elements

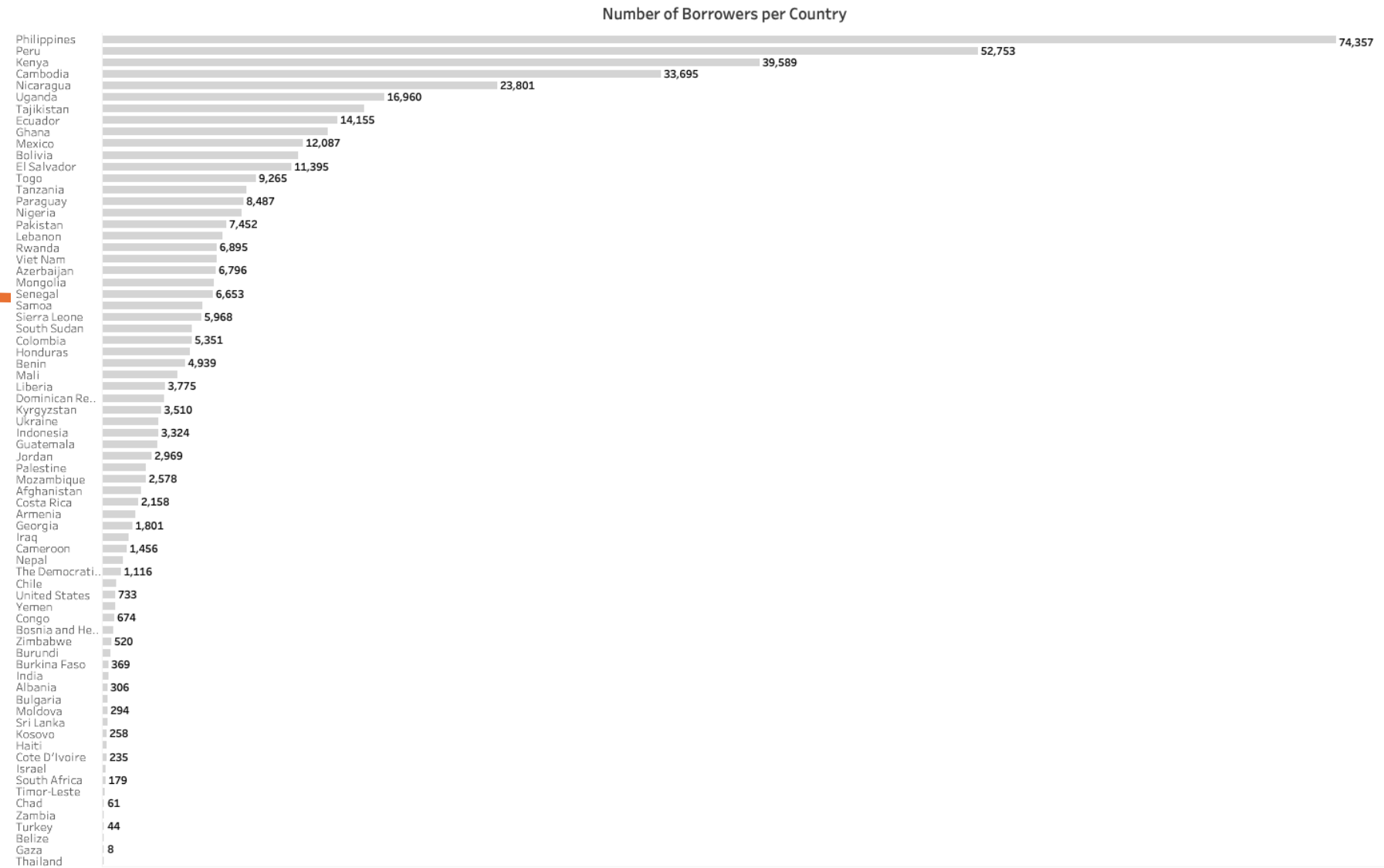
- Difficult to discern specific values
- And understand the distribution and relationships within the data
- Adjust the transparency and/or employ different shapes and sizes



# Too Many Bubbles

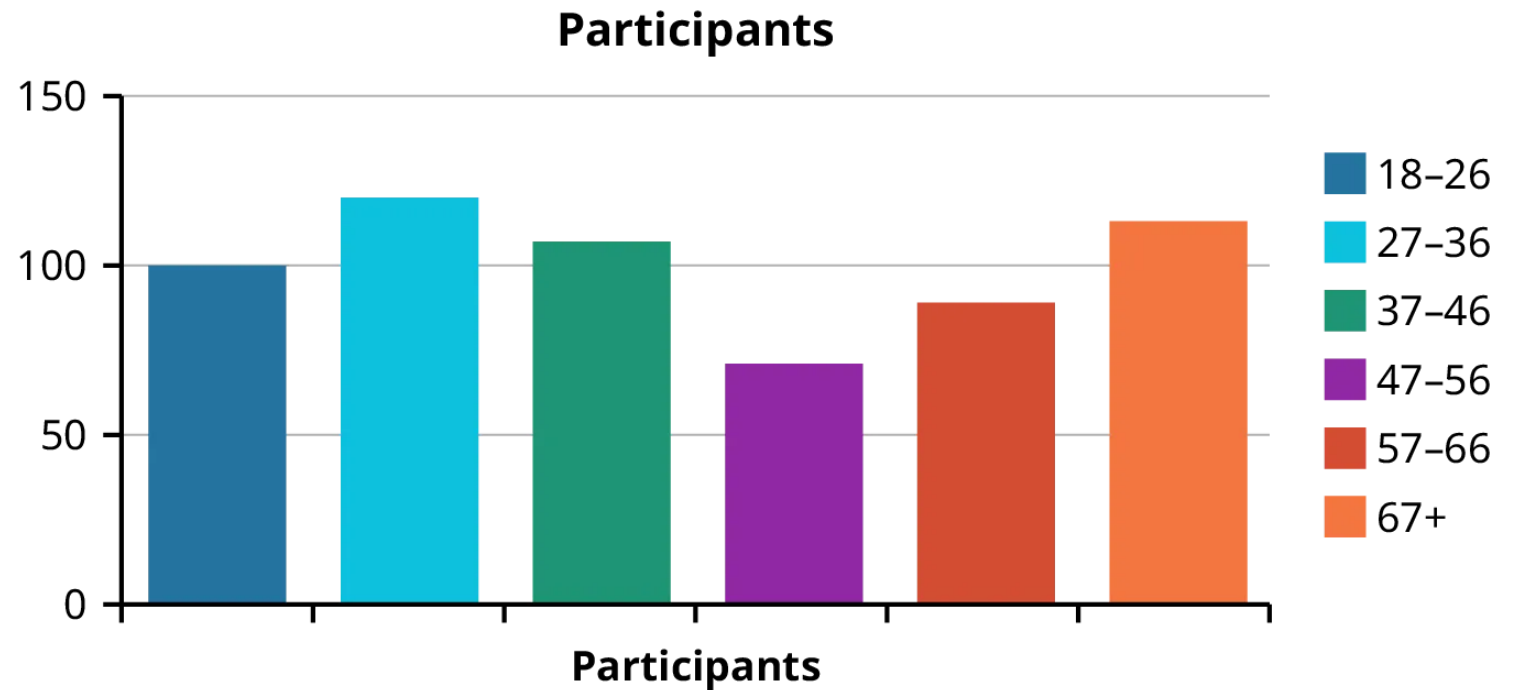


# Better ?



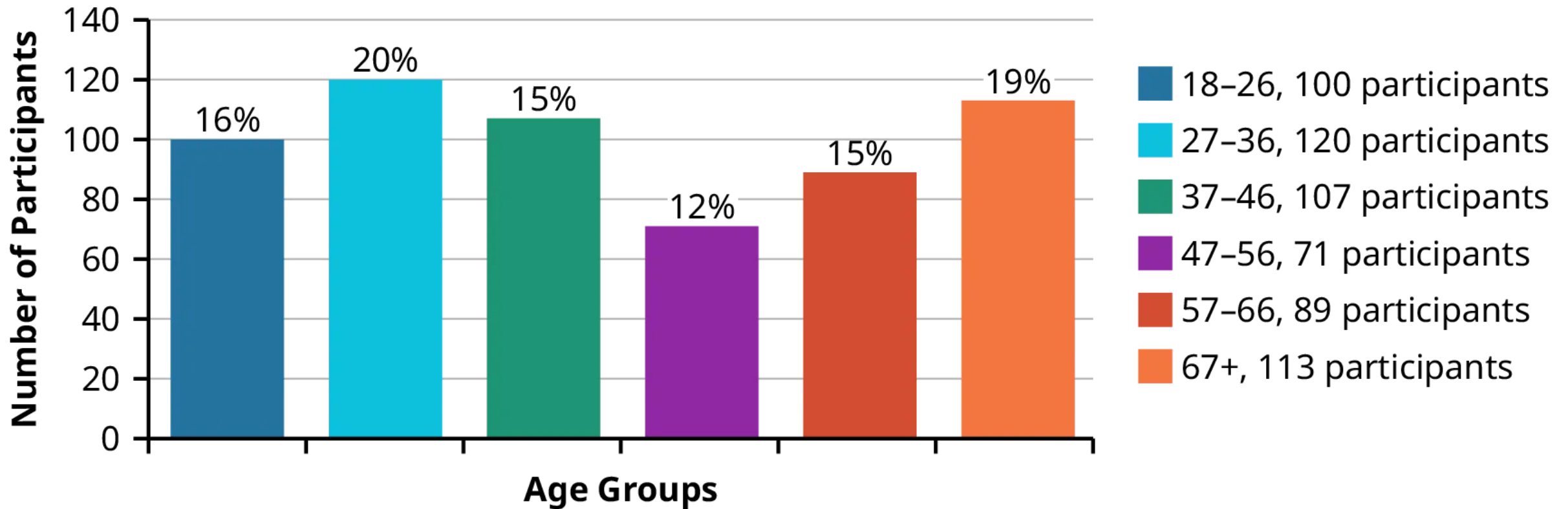
# Missing Information

- Histogram: participants in a clinical trial
- Total number of participants?
- Number of participants per age group?
- Proportion of participants per age group?



# Clear Labeling & Data Information

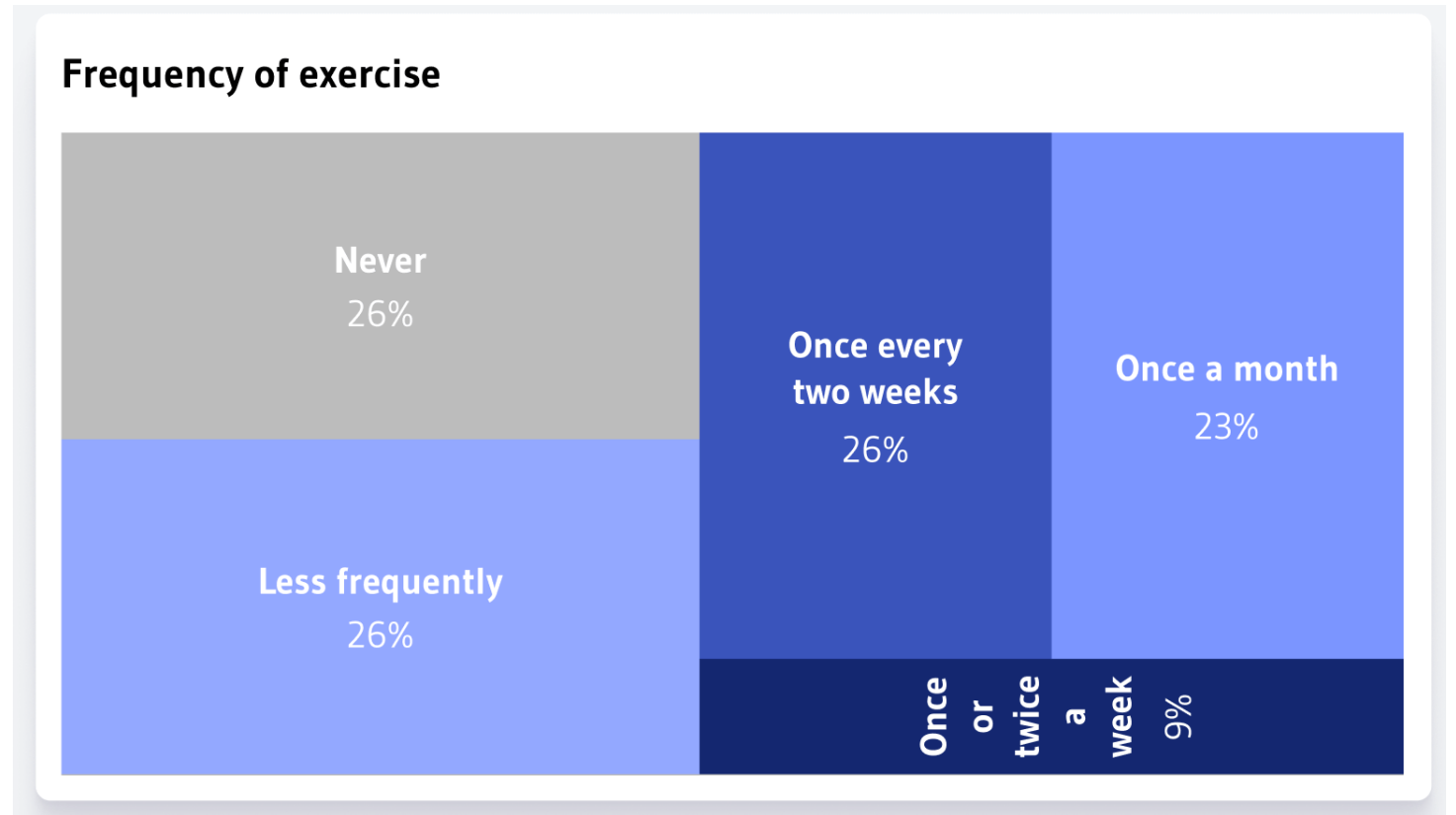
**500 Participants**



# Neglecting to Tell a Story

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- What are the key insights or conclusions ?
- Viewers fail to understand the main message !
- Highlight the most important data points
- Organize the visuals in a way that guides the viewer



# Further Reading



# References

- <https://profspeak.com/data-visualization-gone-wrong/>
- <https://www.gooddata.com/blog/bad-data-visualization-examples-that-you-can-learn-from/>
- <https://www.thoughtspot.com/data-trends/data-visualization/how-to-identify-misleading-graphs-and-charts>
- <https://medium.com/@nigelmills2000/the-truth-and-lies-behind-the-infamous-blood-graph-f6d6691c3626>
- <https://futurism.com/correlation-vs-causation-2>
- <https://www.displayr.com/what-is-overplotting/>
- <https://www.datylon.com/blog/bad-data-visualization-examples>
- <https://www.tableau.com/chart/what-is-bubble-chart>
- <https://openstax.org/books/principles-data-science/pages/8-3-ethics-in-visualization-and-reporting>